GEK-65435

DRIVE SYSTEM INSTRUCTIONS

Directo-Matic® PLUS

FIELD SUPPORT SYSTEM OPERATOR MANUAL

General Electric Company
Drive Systems Department
Salem, Virginia U.S.A.

GENERAL ELECTRIC

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DRIVE SYSTEM INSTRUCTIONS

FIELD SUPPORT SYSTEM

OPERATOR MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

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1. INTRODUCTION TO MANUAL

PURPOSE OF MANUAL

The purpose of this manual is to instruct the DIRECTO-MATIC® Plus control user in Field Support System (FSS) application procedures.

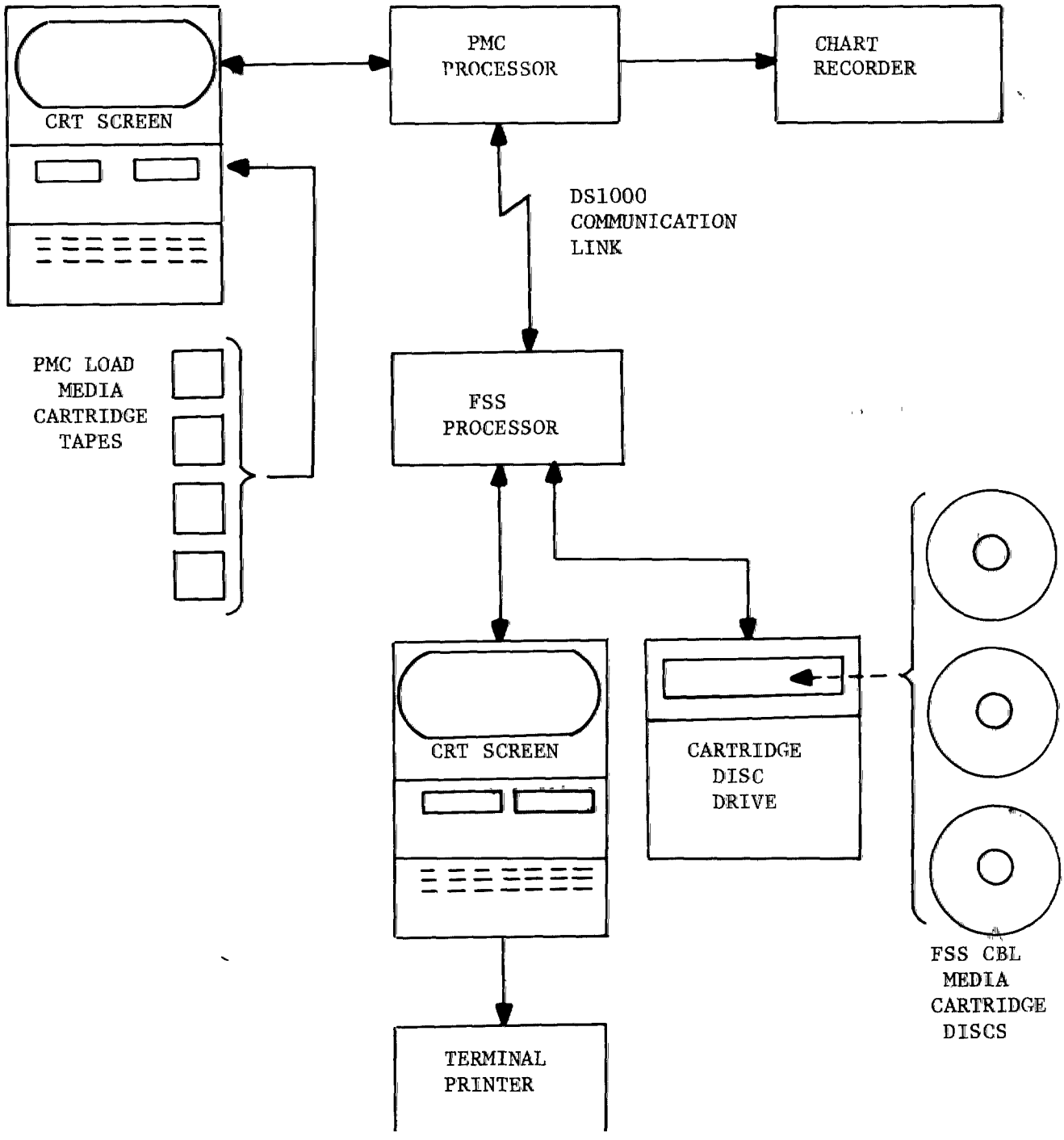
AUDIENCE LEVEL

This publication is directed to experienced electronic technicians. Some experience in programming is preferable but not mandatory.

REFERENCES

DIRECTO-MATIC® Plus Control Programmable Master Controller User's Manual (GEK-65442)

DIRECTO-MATIC® Plus Control Block Language User's Manual (GEK-65426)


CRT SCREEN
PMC
PROCESSOR
CHART
RECORDER
DS1000
COMMUNICATION
LINK
PMC LOAD
MEDIA
CARTRIDGE
TAPES
FSS
PROCESSOR
CRT SCREEN
CARTRIDGE
DISC
DRIVE
FSS CBL
MEDIA
CARTRIDGE
DISCS
TERMINAL
PRINTER

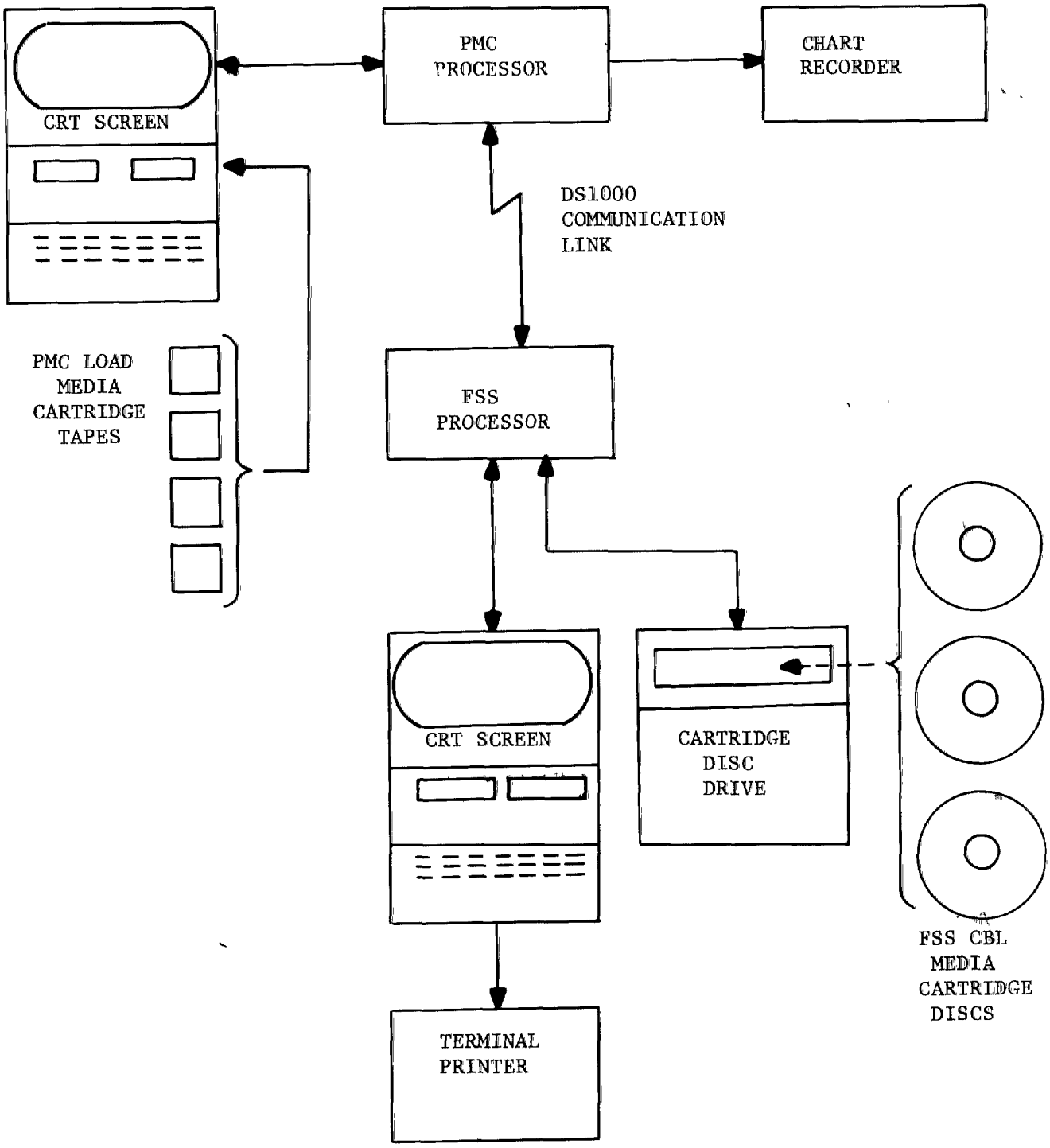


Figure 2-1. Field Support System

2. SYSTEM DESCRIPTION

FUNCTIONAL DESCRIPTION

The Field Support System allows the user to perform the following major functions:

a. Start-up and operate an FSS system.

b, Define and Modify a control block system.

c. Translate control block language (CBL) media to Programmable Master Controller (PMC) load media.

d. Produce documentation of the control block system.

e. Load, copy and compare load media. PMC contents can be loaded from, copied to, or compared to a load media cartridge tape.

f. Monitor and alter the contents of the PMC.

DESCRIPTION OF EQUIPMENT

A block diagram of the FSS system is shown in Figure 2-1.

The FSS processor is programmed to carry out field support functions. There is an interface linking the FSS processor to a PMC. Information can be moved to and from the PMC.

Signals may be trended on a chart recorder via an Analog Signal Segment, driven directly from the PMC.

The user interface is a terminal with CRT display (See Figure 2-2), a cartridge tape drive and keyboard.

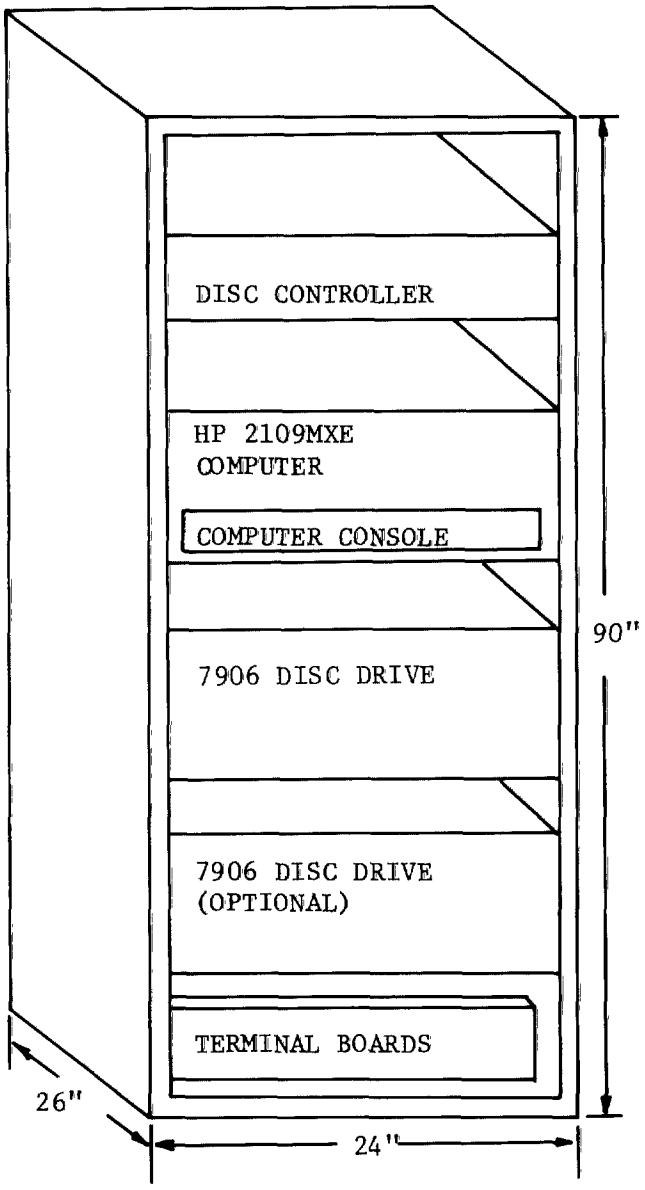
The operator communicates by inputting data through the keyboard and viewing the results on the CRT screen. The cartridge tape is used for storing PMC information. Data cartridge tape can be moved to and from the PMC.



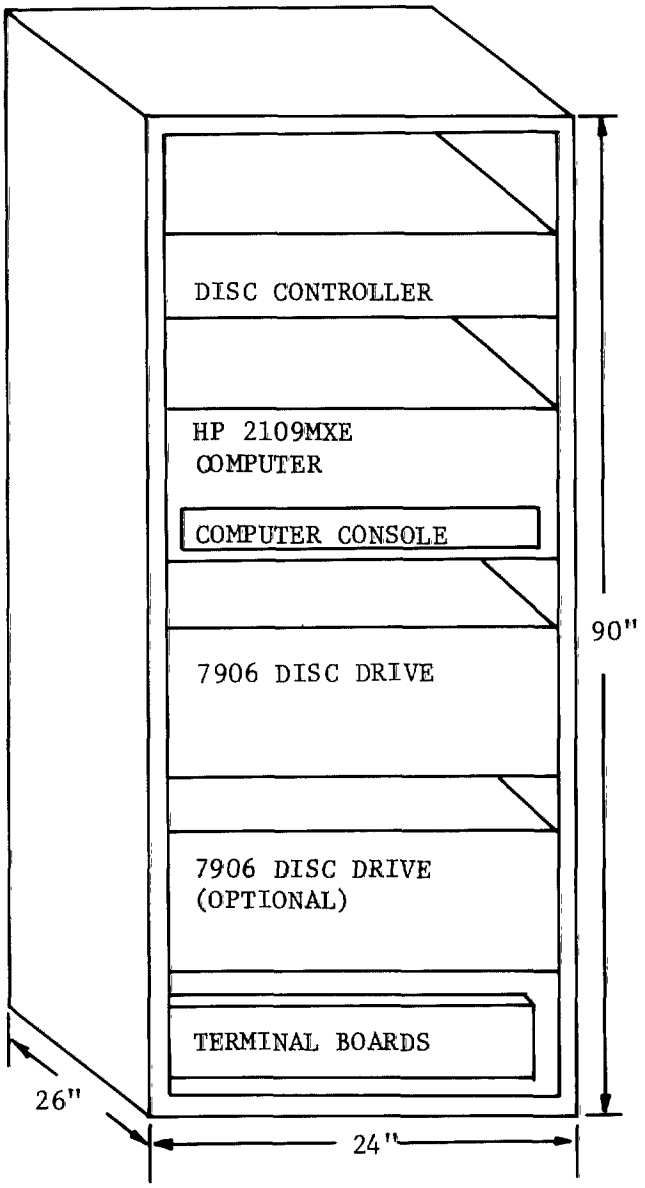
Figure 2-2. Terminal Keyboard and CRT Screen (DSF52648)

The cartridge disc is used for storing CBL media. The FSS processor (See Figure 2-3) accesses the cartridge disc during control system definition and modification.

The terminal printer is necessary to document the PMC control block system.



NOTE: (Cabinet layout will vary if FSS is provided with a computer that performs process automation functions.



NOTE: (Cabinet layout will vary if FSS is provided with a computer that performs process automation functions.

Figure 2-3. DS3840FSS Processor

DOCUMENTATION

This manual details the commands necessary to carry out the functions outlined under the Functional Description.

It is assumed that the reader is familiar with DIRECTO-MATIC Plus Control concepts and the Control Block Language. However, the Control Block Language Reference Manual (GEK-65426) is a useful reference document.

3. OPERATING INSTRUCTIONS

INITIAL FSS START-UP PROCEDURE (Refer to Figure 2-2 for steps (1) thru (9)).

(1) Connect interface cable between the FSS and the HP2645A terminal.

(2) Set terminal POWER switch, on back of terminal, to ON. After approximately 15 seconds, the message TERMINAL READY will appear on the terminal CRT and the cursor will be blinking.

(3) Set terminal HALF/FULL DUPLEX switch to FULL position.

(4) Set terminal EVEN/ODD/NONE PARITY switch to NONE position.

(5) Set terminal BAUD RATE switch to 9600.

(6) Depress terminal REMOTE key into "down" position.

(7) Depress terminal CAPS LOCK key into "down" position.

(8) All other keys should be in "up" position.

(9) Depress RESET TERMINAL key twice in rapid succession. The terminal will beep and a TERMINAL READY message will appear on the CRT. The blinking cursor will appear below the TERMINAL READY message.

(10) Depress the FSS main AC power pushbutton labeled SYSTEM ON/OFF (in upper right-hand corner of FSS cabinet).

(11) Set HP7905A disc drive RUN/STOP switch in STOP position. After the disc drive stops, the DOOR UNLOCKED message will appear on the front of the disc drive.

(12) Select the correct HP12940A disc cartridge(s).

(13) Open the cartridge access door on the disc drive; insert disc cartridge selected in step (12).

(14) Close cartridge access door and set RUN/STOP switch to RUN position. The DOOR UNLOCKED light will go out.

(15) After approximately 5 minutes the DRIVE READY message will appear on the disc drive.

NOTE

Refer to Figure 3-1 for steps (16) thru (23).

(16) Depress the HALT side of the RUN/HALT rocker switch on the front of HP21MX-E computer.

(17) Repeatedly depress either side of the REGISTER SELECT rocker switch until the "S" register red light comes on.

(18) Depress CLEAR DISPLAY side of INSTR STEP/CLEAR DISPLAY rocker switch. All of the display red lights will go out.

(19) Using the 16 DISPLAY REGISTER rocker switches, turn on bit lights, 15, 12, 9, 8 and 1. All other bit lights should be off.

(20) Depress STORE side of STORE/MODE rocker switch.

(21) Depress PRESET side of PRESET/IBL/TEST rocker switch.

(22) Depress IBL/TEST side of PRESET/IBL/TEST rocker switch.

(23) Depress RUN side of RUN/HALT rocker switch. The FSS front panel lights will begin to flash and the disc will be heard working.

(24) The CRT terminal will beep and the following request will appear in a reverse video field on the CRT.

TODAY'S DATE (MONTH, DAY, YEAR) = _

(25) Type in the requested information.

Example: JUN,28, 1979

COMMAS ARE MANDATORY.
TYPING ERRORS MUST BE
CORRECTED BEFORE PROCEEDING
TO THE NEXT STEP

(26) Depress terminal RETURN key.

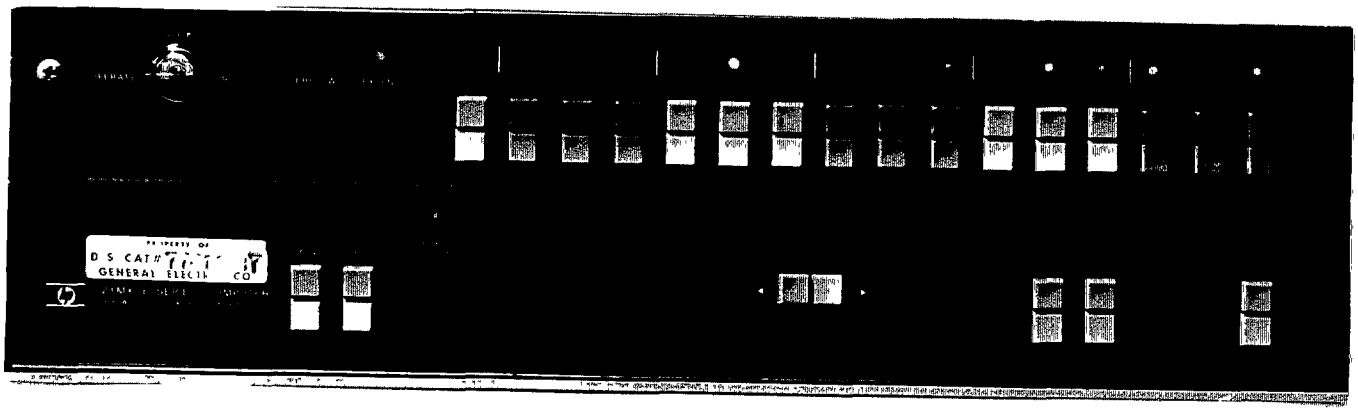
(27) The following request will appear in a reverse video field on the CRT:

NOW SET THE TIME (HOURS, MINS, SECS) _

(28) Type in the requested information.

Example: 8, 40, 30

This is the equivalent of 8 hours, 40 minutes and 30 seconds.



(DSF 52777)

NOTE: All revisions have been upgraded to 2109MXE, Version B which have operational differences in keyswitch.

Figure 3-1, 2113MXE, Version A FSS Front Panel Controls and Indicators.

(29) Depress RETURN key.

(30) The terminal will reply with the time just entered, flashing in reverse video, and will ask: AM OR PM?

8, 40, 30 AM OR PM?

(31) Type in AM or PM, whichever is applicable.

(32) Depress RETURN key. Approximately 55 seconds later the terminal will display the following message:

```
DIRECTO-MATIC PLUS
FIELD SUPPORT SYSTEM
   VERSION N.N
      NNNN
```

Abbreviated Start-Up Procedure (To get reverse video field on CRT)

NOTE

The following assumptions are made:

a. The FSS and the FSS terminal have been initially started-up; otherwise perform the applicable portions of the initial FSS start-up procedure.

b. The desired disc cartridge is running in the disc drive; otherwise, the disc cartridge must be changed (See disc cartridge changing procedure on sheet 3-17).

1. If a colon (:) prompt, followed by the blinking cursor, appears on the FSS terminal, skip steps 2, 3, 4, 5 and 6 and proceed to step 7.

2. If the asterisk (*) prompt, followed by the blinking cursor, appears on the FSS terminal, skip step 3 and proceed to step 4.

3. If only the blinking cursor appears on the FSS terminal with no colon (:) or asterisk (*) preceeding it, depress any key on the terminal keyboard (the letter "K" for example) and the asterisk (*) prompt will appear on the terminal with the cursor blinking next to it.

4. Adjacent to the asterisk (*) prompt, type in:

   RU, FMGØ1 (FMGØ1 is a copy of FMGR-file manager).

5. Depress the RETURN key.

6. The terminal will reply with the colon (:) prompt.

NOTE

If it is desired to execute an audit trail, see Audit Trail procedure, sheet 3-6.

7. Adjacent to the colon prompt, type in:

   :* FSS and depress RETURN key.

   OR

   depress soft key 1 (f1)

   to run FSS.

8. After a few seconds, the following message will appear followed by the reverse video field:

```
  ENTER COMMAND OR SOFT KEY
? [_                                    ]
```

OPERATING INSTRUCTIONS

9. SELECT or CREATE the appropriate control system.

   (Refer to sheet (3-15) for instructions applicable to desired Control System Definition Commands.)

Audit Trails. Audit trail information is contained in Appendix D.

Procedure: To operate the FSS in the audit trail mode, steps 1 through 6 of the abbreviated start-up procedure (See sheet 3-5) should be performed, then enter one of the following commands:

*WRITF (Writes audit trail file)

*READM (Reads audit trail in manual mode)

*READS (Reads audit trail in semi-automatic mode)

*READA (Reads audit trail in automatic mode)

NOTE

The semi-automatic and manual modes ignore the following commands in an audit trail file: STOP, TRANSL, RETRAN, WTAPE, and RTAPE. These modes also append to the audit trail file any operator entries after the audit trail file end-of-file is reached.

NORMAL OPERATION COMMAND FORMAT

All FSS functions are initiated by entering (on terminal keyboard) either a soft key (f1 through f8) or certain control commands (See Table 3-1). A command input is expected whenever the prompt ENTER COMMAND OR SOFT KEY is displayed above the inverse video field (white space) at the top of the CRT. After the FSS is placed in operation, the soft key commands are processed immediately. Table 3-2 serves as a "road map" of soft key commands which enables the operator to select soft keys necessary to arrive at any command function desired.

SOFT KEY MENUS

Soft key menus appear at the lower right hand side of the CRT display. These menus index the programmable soft keys (f1 through f8) on the terminal keyboard. Each soft key allows the selection of a command or menu.

There are eight different sets of menus depending upon the type of function that the operator desires to perform. The MASTER menu contains a list of other menus. When the operator selects one of the other menus by use of a soft key, the menu and soft key definitions change accordingly. By referring to Table 3-2, observe that the editing and interaction menus continue in a cyclical fashion. For example, the editing menu leads to two menus; EDIT CB and EDIT SEG. When in either of these menus, the other menu can be selected by use of soft key f7. This procedure is also applicable to INTERACTION which contains four menus. Soft key f7 allows the operator to sequentially select INTERACT menus A thru D as desired.

SOFT KEY COMMAND HIERARCHY

In the soft key command hierarchy, there are three levels of command. The first or upper level is indicated when the MASTER menu appears in the lower right-hand corner of the CRT. At this time, if any soft key is depressed, the second level (INITIALIZE, EDIT, etc.) will be entered. While at the second level, if a soft key is again depressed, the third and lowest level, e.g. SELECT or CREATE under INITIALIZE menu is entered. An operator at the third level desiring to return to the second level may depress the terminator (ENTER) twoce. An operator at the second level desiring to return to the first level needs only to depress the f8 soft key.

An inexperienced operator should be assisted in his initial attempts at using the soft key commands by referring to "A Typical Editing Session Using the FSS" in Section 4.

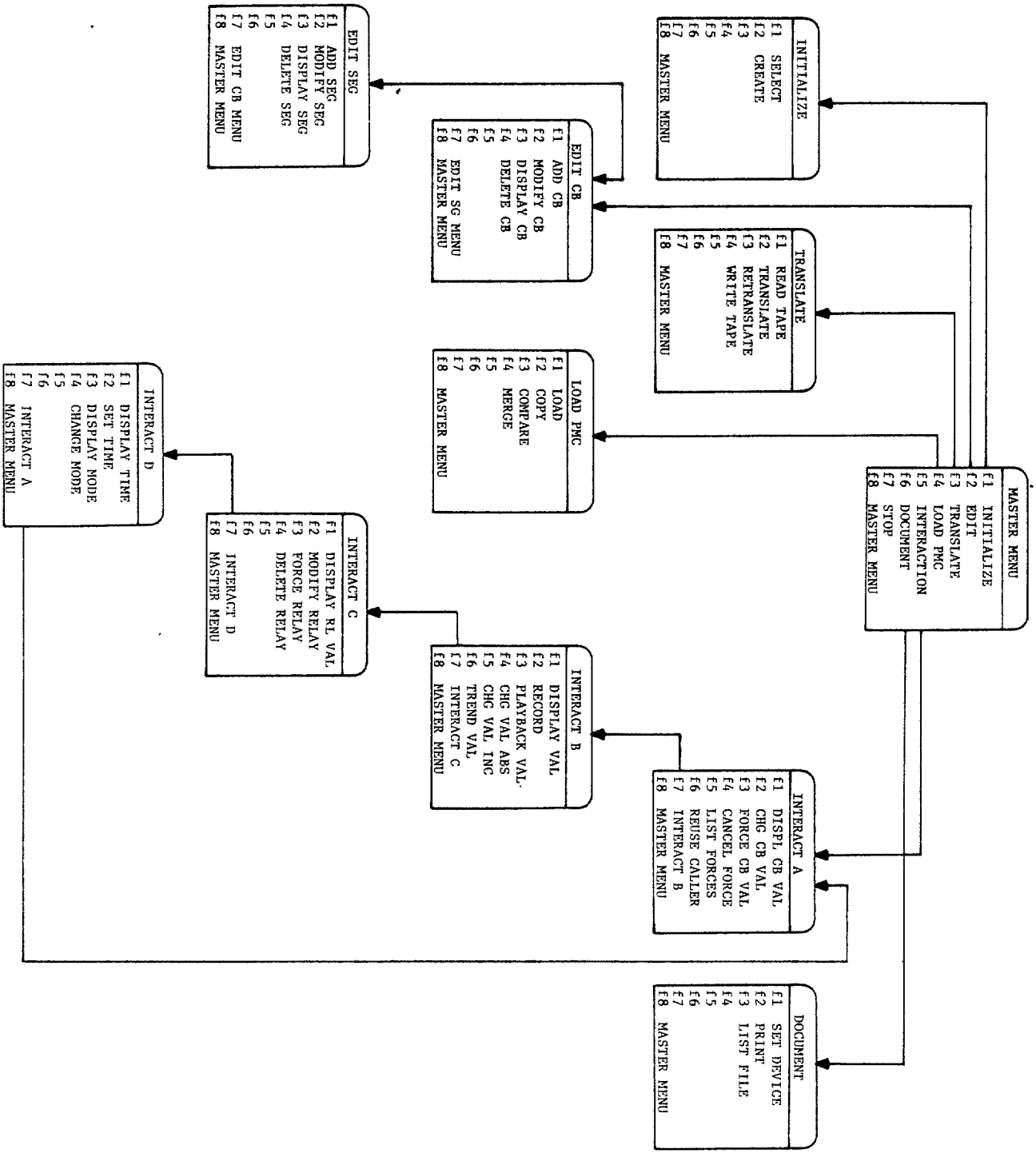
TABLE 3-1. MENU AND CONTROL COMMANDS

MENU	SOFT KEY	MENU NAME	CONTROL COMMAND	SHEET NO
MASTER				
	f1	INITIALIZE		
	f2	EDIT		
	f3	TRANSLATE		
	f4	LOAD PMC		
	f5	INTERACTION		
	f6	DOCUMENT		
	f7	STOP	STOP	3-16
	f8	MASTER MENU		
INITIALIZE				
	f1	SELECT	MNTFD	3-18
	f2	CREATE	INIT	3-19
	f8	MASTER MENU		
EDIT CB	f1	ADD CB	ADDCB	3-21
	f2	MODIFY CB	MODCB	3-26
	f3	DISPLAY CB	DISPCB	3-27
	f4	DELETE CB	DELCB	3-28
	f7	EDIT SG MENU		
	f8	MASTER MENU		
EDIT SEG				
	f1	ADD SEG	ADDID	3-29
	f2	MODIFY SEG	MODID	3-32
	f3	DISPLAY SEG	DISPID	3-33
	f4	DELETE SEG	DELSD	3-34, 3-35
	f7	EDIT CB MENU		
	f8	MASTER MENU		
TRANSLATE				
	f1	READ TAPE	RTAPE	3-36
	f2	TRANSLATE	TRANSL	3-37
	f3	RETRANSLATE	RETRAN	3-38
	f4	WRITE TAPE	WTAPE	3-39
	f8	MASTER MENU		

TABLE 3-1. MENU AND CONTROL COMMANDS (CONT.)

MENU	SOFT KEY	MENU NAME	CONTROL COMMAND	SHEET NO
LOAD PMC				
	f1	LOAD	FLOAD	3-41
	f2	COPY	COPY	3-42
	f3	COMPARE	COMPAR	3-44
	f4	MERGE	MERGE	3-36
	f8	MASTER MENU		
INTERACT A				
	f1	DISPL CB VAL	DSPCB	3-48
	f2	CHG CB VAL	CHANGE	3-50
	f3	FORCE CB VAL	FORCE	3-52
	f4	CANCEL FORCE	CANCEL	3-54
	f5	LIST FORCES	LISTFR	3-55
	f6	REUSE CALLER	CALLER	3-56
	f7	INTERACT B		
	f8	MASTER MENU		
INTERACT B				
	f1	DISPLAY VAL	DISPV	3-57
	f2	RECORD VAL	RECORD	3-59
	f3	PLAYBACK VAL	PLAY	3-61
	f4	CHG VAL ABS	CHGABS	3-63
	f5	CHG VAL INC	CHGINC	3-65
	f6	TREND VAL	TREND	3-67
	f7	INTERACT C		
	f8	MASTER MENU		
INTERACT C				
	f1	DISPLAY RL VAL	DSPLD	3-68
	f2	MODIFY RELAY	MDFLD	3-69
	f3	FORCE RELAY	FRCLD	3-71
	f4	DELETE RELAY	DLTLD	3-72
	f7	INTERACT D		
	f8	MASTER MENU		
INTERACT D				
	f1	DISPLAY TIME	DTIME	3-73
	f2	SET TIME	STIME	3-74
	f3	DISPLAY MODE	DMOD	3-75
	f4	CHANGE MODE	CMOD	3-76
	f7	INTERACT A		
	f8	MASTER MENU		
DOCUMENT				
	f1	SET DEVICE	SETDD	3-77
	f3	LIST FILE	LIST	3-80
	f8	MASTER MENU		

TABLE 3-2. SOFT KEY COMMAND HIERARCHY



SOFT KEY COMMAND OPERATION

To execute a soft key command, proceed as follows:

1. Depress soft key (f1 through f8) corresponding to a command appearing on the menu. The command name appearing on the menu will blink.

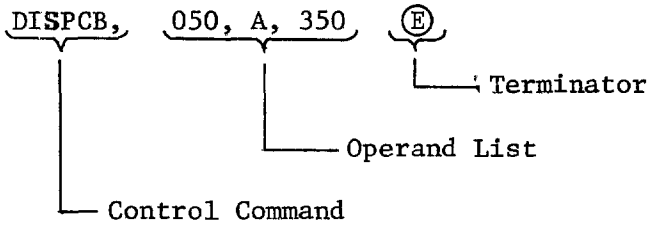
2. A prompt of the operand list for the selected command will appear above the inverse video field on the CRT. Optional operands appear in parentheses. Key in the desired operands.

3. Depress ENTER (Ⓔ) key.

CONTROL COMMANDS

Some of the soft key commands have equivalent "Control Commands" as shown in Table 3-1. A control command must be keyed-in character-by-character as viewed in the inverse video field on the CRT. After the control command and supporting information has been keyed in a terminate action is necessary. This is accomplished by depressing the ENTER key (Ⓔ). This signals the FSS to interpret the command and carry out the requested function.

All control commands have the following general format:



- Command Name
  - alphabetic
  - identifies function
- Operand List
  - alpha numeric
  - comma-separated, variable number of operands
- Terminator
  - normally the ENTER, designated by Ⓔ.
  - occasionally the RETURN, designated by Ⓒ,

Commands shown in this publication follow certain conventions illustrated below:

DISPCB, blnb, seg, drive (E)

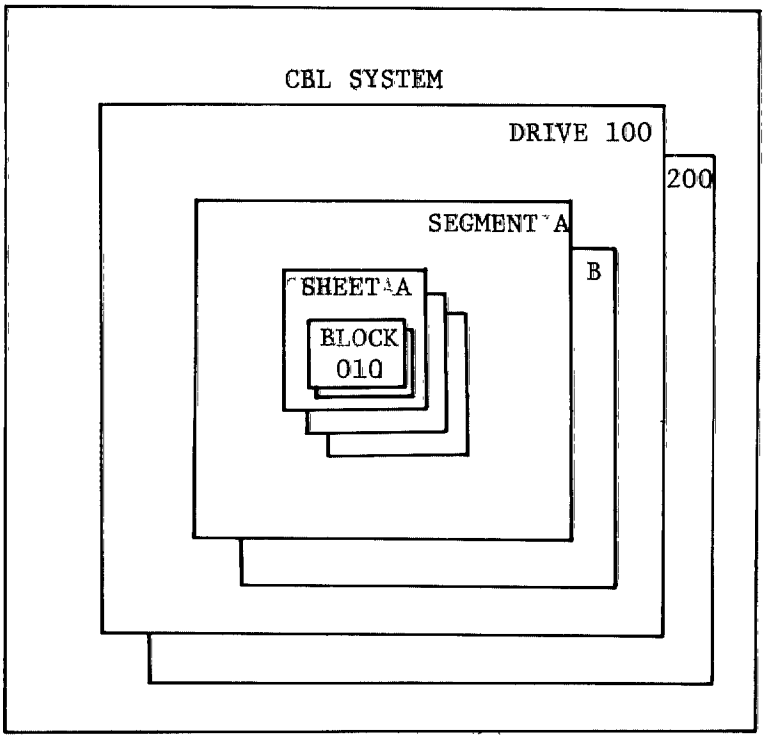
- The command name is shown in upper case letters and must be entered exactly as shown.
- The operands are variables shown in lower case. They are to be replaced by values as defined in the text.
- There is no comma between the last operand and terminator.
- Operand leading and trailing blanks are ignored.

When a command is entered, one of three results will occur;

- Command accepted and executed.
- Further data required and the operator is prompted to enter it.
- An error is encountered and an error message displayed.

CONTROL SYSTEM DEFINITION COMMANDS

The elementary drawings of a control system are in the form of control blocks and their interconnections. The basic drawing is a sheet. Each sheet is a size B (11 x 17 inches) drawing. The next higher drawing level is a segment containing one or more sheets. The next level is a drive, which contains one or more segments. The CBL system, the highest level, is one or more drives containing a complete set of drawings. See CBL system structure below:



Implementation of the elementary drawings is through control block commands which create a CBL system or modify a previously created system. A translation is implemented to produce load media for the PMC.

To create a new control block system, the PMC to be used must be identified. A cartridge disc is mounted and the "CREATE" command is performed.

To modify an existing system, the PMC CBL media cartridge disc is installed and the "SELECT" command is used to verify controller identity.

Segment identification information is associated with each segment of a system. A group of menu names or commands is provided to display, add or modify a segment title, revision level and revision date, plus a number of parameters used during segment execution.

A control block may be displayed, added to the system, modified or deleted from the system. A separate menu name or command exists for each of these operations.

Each reference to a control block must include drive, segment and sheet identification which is included as operands in the control block commands.

Once entered, the drive, segment and sheet numbers are retained until superceded. Control blocks which are part of the same segment and drive can be manipulated without entering drive and segment numbers again. Short format commands are provided for this. When defining another drive and/or segment, the new numbers must be entered using the longer format commands.

Each control block command entry will display the requested block on the CRT. Refer to CBL Reference Manual GEK-65426 for descriptions of these displays and their input/output signals.

NOTE

The abort key (f1) can be used in control block editing commands to suspend the command if desired.

If a control block is being added or modified, the cursor must be pre-positioned at the beginning of an automatic sequence of inverse video fields for the particular block. The appropriate value should be keyed into the correct position within the block.

Depressing the TAB key moves the cursor to the next data input position. Depressing the TAB key while the CONTROL key is held down moves the cursor to the previous input position. If no value is to be entered, only the TAB key needs to be depressed to skip the entry. The sequence of inputting data is continued until all information for the control block has been entered and the ENTER key is depressed.

Adding or modifying relay ladders is slightly different from control block editing. Relays have special menus which are used for editing. The tab key should not be used. Refer to the section on relay ladders for further details.

Whenever a data field is being modified, the remnants of right-justified data must be spaced out to properly clear old characters. Depress the ENTER key to terminate the input sequence.

Menu and control command procedures are detailed on the sheets indicated in Table 3-1.

Command operands are listed and identified in Appendix B.

DISC PACK CHANGING PROCEDURE

NOTE

Multiple control systems can be used, each one using two cartridges on a disc pack.

The following procedure will transfer you from the FSS mode (ENTER COMMAND message followed by a reverse video field) using one system's disc to the FSS mode using the other system's disc.

1. Perform STOP procedure described in Section 3.

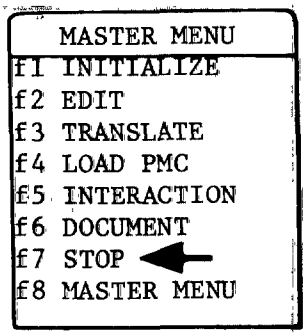
2. Set the RUN/STOP toggle switch on the HP7905A or HP7906 disc drive to STOP position. The DRIVE READY light will go out.

3. After the disc drive stops, the DOOR UNLOCKED message will appear on the front of the disc drive panel.

4. Open the cartridge access door on the disc drive and remove the disc pack.

5. Place the disc in the disc storage rack inside of the storage cabinet.

6. Now perform the initial FSS start-up procedure on page 3-1 beginning with step 12.



STOP

Function:

This command initiates an orderly shutdown of the FSS. When the FSS is not being used it should be left in a quiescent state. This will avoid electrical or physical damage to the disc cartridge in the disc drive in the event of an electrical power interruption or an accidental turn-off of the disc drive.

Format: f7 (STOP) (E)

PROCEDURE:

1. If the last display on the FSS terminal is a blinking cursor only (no reverse video field, no colon (:) prompt, or no asterisk (*) prompt present), no further action is necessary.

2. If the last display on the FSS terminal is an asterisk (*) prompt (no reverse video field or no colon (:) prompt present), skip steps 3 and 4 and proceed to step 5 below.

3. If the last display on the FSS terminal is a colon (:) prompt, skip step 4 and proceed to step 5 below.

4. If the last display on the FSS terminal is:

```
ENTER COMMAND OR SOFT KEY

? _
```

Soft Key Method:

Depress f8 (MASTER MENU), then depress f7 (STOP).

Result:

```
FSS SHUTDOWN COMPLETE
***PRESS SOFT KEY 1 (f1) TO RUN FSS***
PRESS SOFT KEY 2 (f2) TO RUN *WRITF
```

Alternate Method:

Type the command STOP into the reverse video field as follows:

```
ENTER COMMAND OR SOFT KEY

? STOP
```

Depress ENTER key.

Result: This disc will be heard working and the terminal will reply:

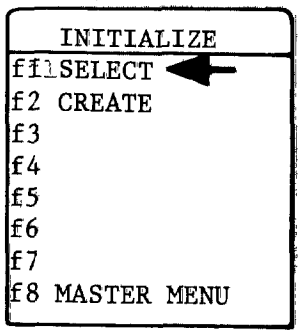
```
FSS SHUTDOWN COMPLETE
***PRESS SOFT KEY 1 (f1) TO RUN FSS***
PRESS SOFT KEY 2 (f2) TO RUN *WRITF
```

5. After a colon (:) appears, depress the RETURN key.

6. The following message will appear:

```
:END FMGR
```

7. This completes the normal shutdown of the FSS terminal.



SELECT SYSTEM

Function: This command allows a particular PMC control system to be prepared for access. The command must be executed before any edit or interaction commands can be performed. The FSS data base file (stored on cartridge disc) must be created prior to this command, otherwise a -6, FILE ACCESS ERROR will occur.

NOTE

Ensure applicable disc is mounted prior to performing following procedure.

Format:

f1(SELECT)nnnn(system ID)(E)

Procedure:

1. Depress f1 (SELECT)key on 'INITIALIZE menu.'

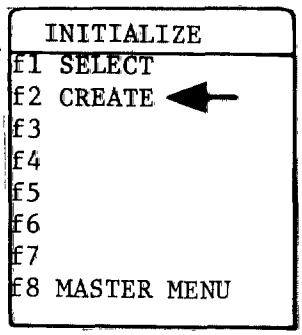
   The following prompt will appear:

   ENTER ID

2. Key in nnnn (system ID number.)
3. Depress (E).

Result:

System nnnn modification can begin.



SYSTEM INITIALIZATION

Function: Initializes the system for editing and interaction commands. This command is used to initialize the data base for a particular control system. All data previously entered for the specified system is deleted.

Format:

f2 (CREATE)nnnn(system ID)Ⓔ

Procedure:

1. Depress f2(CREATE)key on INITIALIZE menu.

   The following prompt will appear:

   ENTER ID

2. Key in nnnn(system ID number)
3. Depress Ⓔ.

Result:

If the system nnnn was previously created, the following prompt will appear:

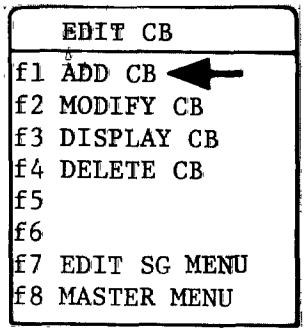
IT IS OK TO DELETE FILES Y/N?



This command destroys any information previously recorded on the identified system.

Y - Should be keyed in if you wish to delete the previously entered system, then depress (E).

N - Should be keyed in if you wish to retain the previously entered system, then depress (E). No action will result. Depress f1 (SELECT) instead of f2 (CREATE) to avoid system deletion.



ADD CONTROL BLOCK

Function: This command is used to add a control block to an existing system.

Format:

f1(ADD CB) aaaa (type*), nnn (block number), a (sheet), a (segment), nnn (drive) (E)

Procedure:

1. Depress f1 (ADD CB) key on (EDIT CB menu.)

   The following prompt will appear:

   ENTER TYPE, BLOCK # (, SHEET(,SEG(,DRIVE))

2. Key in aaaa (type*), nnn (block number), a (sheet), a (segment), and nnn (drive) separated by 4 commas.

3. Depress (E).

4. If any of the parameters were invalid, error messages will appear and the improper parameters may be re-entered.

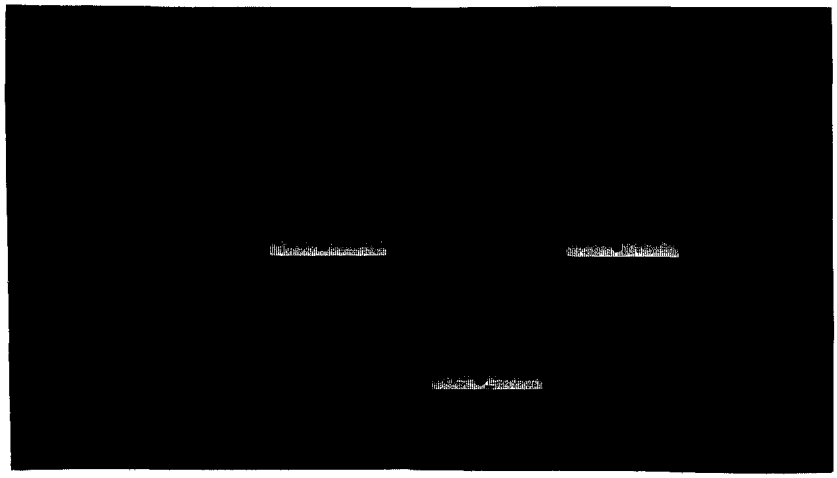
5. When all parameters are correct, a block picture will appear.

6. The method of completing the block addition will depend on the block type chosen. If a relay block were chosen (KRLY, KCTR, KDLY, KLCH, KPLS, or KHLD), go to step 11.

NORMAL CONTROL BLOCK EDITING

7. For example, in response to:
AABS,10,Z,A,1 ⓔ

the screen will display the following picture:



NOTE

Adding a block may be aborted at any time by pressing the abort key (f1).

Key the applicable information into the inverse fields, using the tab key to move the cursor to each field. Definitions for the fields are explained for each block in the CBL Reference Manual (GEK-65426).

8. Press ⓔ when editing is complete.

9. If the syntax or type of signals used in the fields is improper, an error message will appear and the block may be re-edited as in step 7 and 8.

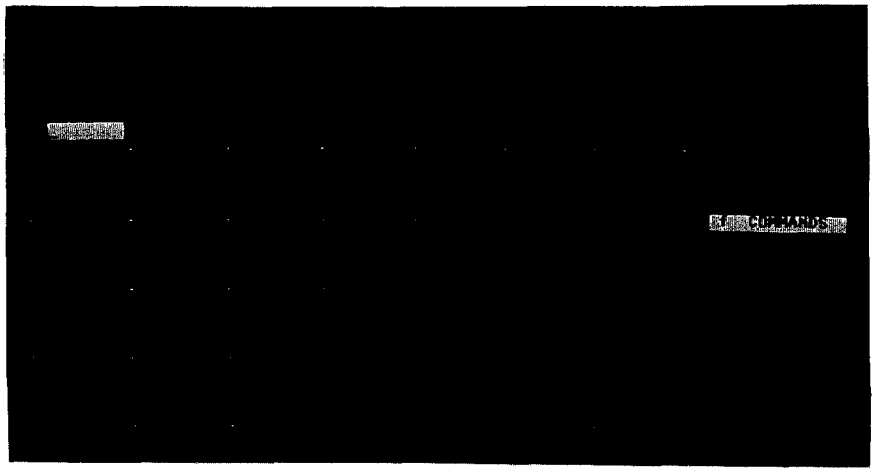
10. When all fields are completed properly, the block will be entered into the system and another command may be selected from the menu.

RELAY BLOCK EDITING

11. For example, in response to:

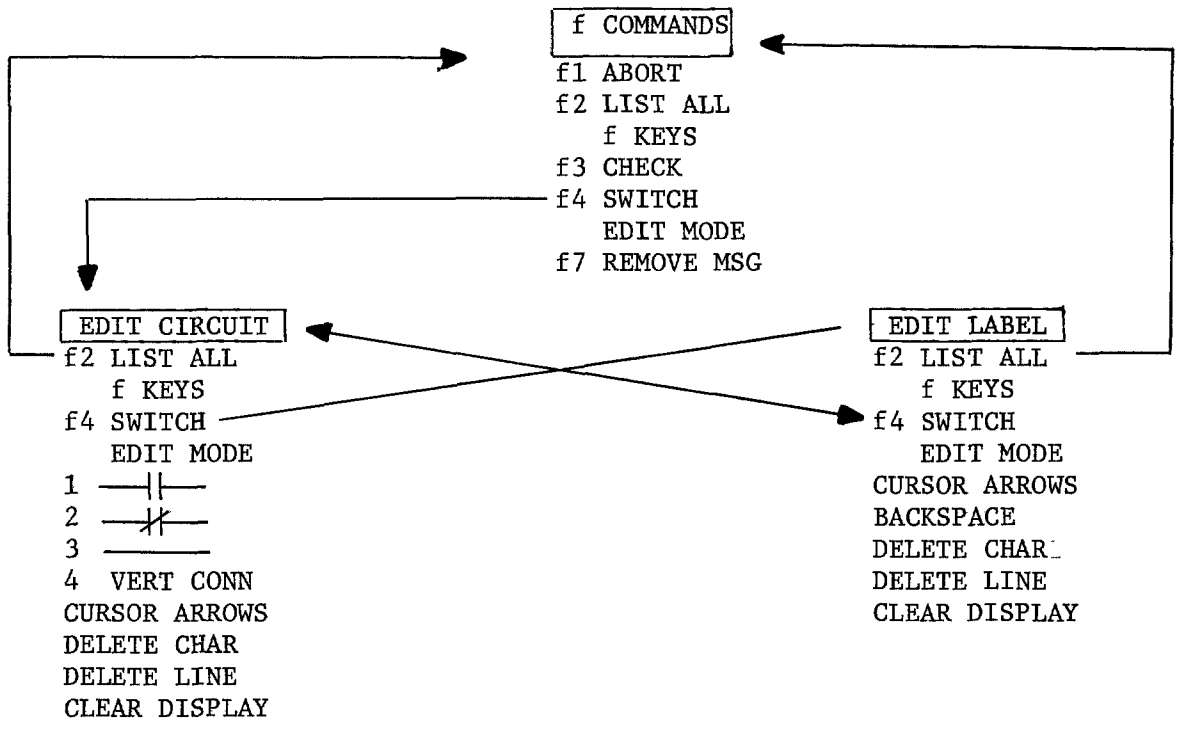
KRLY,10,A,Z,10 ⓔ

the screen will display the following picture:



and a new menu would appear. This menu and the 2 other menus in this group are shown below.

RELAY LADDER EDITING MENUS



NOTE

Addition of a block may be aborted at any time by pressing the abort key (f1).

Information about the relay blocks may be found in the Control Block Language Reference Manual (GEK-65426).

12. Press the f4 key to enter the Edit Circuit menu.

13. Circuit editing can only be performed on the KRLY, KCTR, and KDLY blocks. If a KHLD, KLCH, or KPLS block is chosen, go to step 16. Circuit editing may be performed on relay blocks as listed:

KRLY	all circuit elements
KCTR	input contact elements
KDLY	time delay value only

14. Modify the circuit by using the commands indicated in the menu.

    a) MOVING THE CURSOR. Press the cursor arrows (left, right, up, down, and home) to move the cursor around the circuit elements. The "return" key moves the cursor to the left-most field in the present line.

    b) CHANGING THE CIRCUIT ELEMENTS. Press the numeric keys 1-4 to modify a circuit element on which the cursor is positioned.

    c) DELETING CIRCUIT ELEMENTS. Use the following keys to erase portions of the circuit.

delete char -	Erases the element on which the cursor is positioned
delete line -	Erases the line on which the cursor is positioned
clear display -	Erases the remaining portion of the row to the right of the cursor position and erases all rows below the cursor position

15. Special Notes for KRLY Editing:

    - NORMALIZATION: When a "check" key is pressed, the correctness of the circuit will be verified, and the circuit elements will be repositioned to the top-most, left-most position possible without changing the character of the circuit.

    - WRAPAROUND: Within the 5x7 relay ladder diagram it is possible to create relay strings with greater than 7 contacts in series.

15. Continued

A maximum of 19 series contacts are possible. To extend a relay ladder row beyond 7 series contacts the operator extends a relay row by connecting its end to the start of another row which is used as a shorting or "wraparound" row. A maximum of two wraparounds are possible which yields 19 series contacts. Note that in this configuration no "or-ing" is possible. The "shorting" or "wraparound" row cannot be intersected.

- OR-ING: A maximum of 5 OR'd relay ladder rungs is permitted, although it is physically possible to create relay ladders with greater than 5 OR'd rings within the 5 x 7 format. If the limit is exceeded, an error message will be displayed when normalization is attempted subsequent to pressing the "enter" or "check" keys.

16. When the circuit is in the desired form, press the f4 key to edit the label fields.

17. Modify the labels using the commands indicated by the menu.

a) MOVING THE CURSOR. Press the cursor arrows (left, right, up, down, and home) and "return" to move the cursor to different label fields. Only the label field on which the cursor is positioned will appear in inverse video.

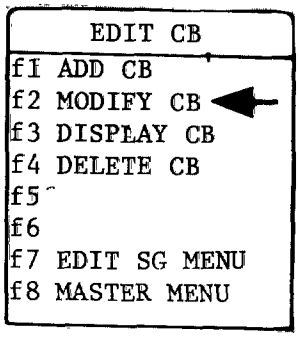
b) CHANGING A LABEL. After positioning to the desired location, type in the logical signal name or signal number desired. The backspace key will move the cursor left without affecting the characters "backed over". The space key will blank out a character.

c) DELETING A LABEL FIELD. Press the delete character key to erase the label field on which the cursor is positioned.

18. When the circuit is complete, press ⓔ.

19. Normalization and verification will be performed. If an error is found, a message is displayed and the relay block is redisplayed. Press f7 to remove the error message from the screen. The block may be re-edited as in steps 12-18.

20. When the circuit is completed with no errors, the relay block will be entered into the system and another command may be selected from the menu.



MODIFY CONTROL BLOCK

Function: This command is used to modify a previously entered control block.

Format:

f2 (MODIFY CB) nnn (block number), a (segment), nnn (drive) (E)

Procedure:

1. Depress f2(MODIFY)CB key on EDIT CB menu.

   The following prompt will appear:

   ENTER BLOCK # (,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

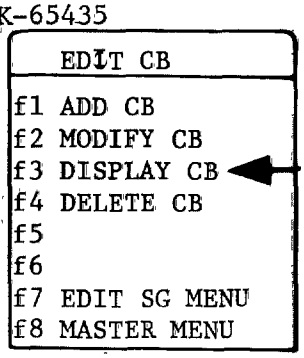
3. Depress (E).

Result:

A previously entered control block will appear on the CRT screen. All of the previous responses to the inverse video fields will be displayed. All of these fields may now be edited in the same manner as when the original block was entered using the ADD CONTROL BLOCK command. Detailed information regarding editing is contained in the explanation of ADD CONTROL BLOCK beginning on page 3-21.

NOTE

The block number cannot be changed by use of the MODIFY CONTROL BLOCK command. To change the block number, a DELETE CONTROL BLOCK command must be executed, followed by an ADD CONTROL BLOCK command.



DISPLAY CONTROL BLOCK

Function: This command is used to display a previously assigned control block on the CRT screen or printer.

Format:

f3 (DISPLAY CB) nnn (block number), a (segment), nnn (drive) (E)

Procedure:

1. Depress f3(DISPLAY) CB key on EDIT CB menu.

   . The following prompt will appear:

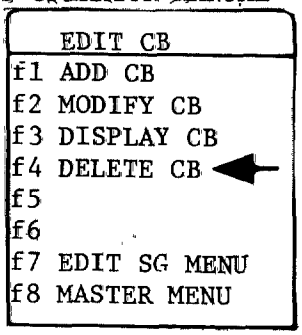
   ENTER BLOCK # (,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E)

Result:

The identified control block will appear on the CRT screen or printer (refer to SET DISPLAY DEVICE command.



DELETE CONTROL BLOCK

Function: This command is used to delete a previously entered control block from a designated segment/drive.

Format:

f4 (DELETE CB) nnn (block number), a (segment), nnn (drive) (E)

Procedure:

1. Depress f4(DELETE CB)key on EDIT CB menu.

   The following prompt will appear:

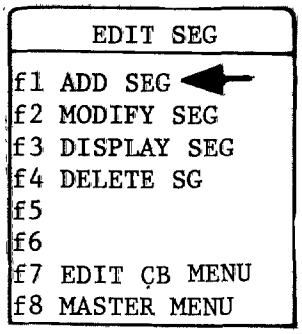
   ENTER BLOCK # (,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

Result:

The control block will be deleted from the designated segment/drive.



ADD SEGMENT IDENTIFICATION

Function: This command is used for the initial entry of segment identification information. If segment identification has been previously input for this segment/drive and a change is desired, the MODIFY SEGMENT IDENTIFICATION function must be used.

Format:

f1 (ADD SEG) a (segment), nnn (drive) Ⓔ

Procedure:

1. Depress f1(ADD SEG)key on EDIT SEG menu.

   The following prompt will appear:

   ENTER SEG (,DRIVE)

2. Key in a (segment), and nnn (drive) separated by a coma.

3. Depress Ⓔ.

Result:

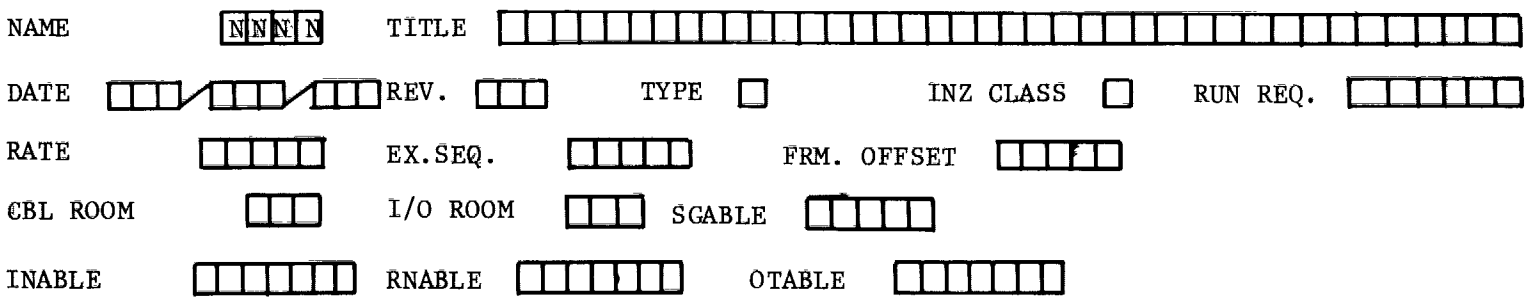
Segment identification block appears.

(See Table 3-3 for a definition of segment identification input data).

Use TAB key to move cursor to each of the modifiable fields. Key in data changes and/or additions. Depress Ⓔ when editing is completed. If the syntax used in the fields is improper, an error message will appear and the segment identification may be re-edited to correct the error.

The abort key (f1) may be used to abort the editing if desired.

SEGMENT IDENTIFICATION



Refer to page 3-29 for a definition of input data items.

Figure 3-2. Segment Identification Format

TABLE 3-3. SEGMENT IDENTIFICATION INPUT DATA

TITLE - Alpha-numeric description of the segment.

DATE - Current revision month, day, year.

REV. - Numeric revision level.*

● TYPE - Segment type. P - procedural
C - continuous
I - interrupt
Z - initialize

INZ CLASS - Initialize segment type for initialize segments only.
H - hot start
C - cold start

RUN REQ. - Name of run request logical variable for procedure segment only.

RATE - Numeric segment repetition rate in multiples of the time frame rate

EX.SEQ - Numeric segment priority number.

FRM,OFFSET - Numeric segment execution time offset in multiples of the time frame rate

CBL ROOM - Percentage spare room in the segment

I/O ROOM - Percentage spare I/O control table room

● SGABLE - Name of the segment enable logic variable

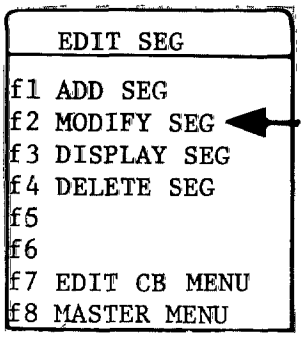
● INABLE - Name of the input phase enable logic variable

● RNABLE - Name of the run phase enable logic variable

● OTABLE - Name of the output phase enable logic variable

*If the revision level is set to #99 the segment will not be translated. This enables the user to specify which segments are to be translated for testing in a PMC.

● Required inputs



MODIFY SEGMENT IDENTIFICATION

Function: This command is used to modify previously entered segment identification information. Missing data can also be entered.

Format:

f2 (MODIFY SEG) a (segment), nnn (drive) Ⓔ

Procedure:

1. Depress f2 MODIFY SEG key on EDIT SEG menu.

   The following prompt will appear:

   ENTER SEG (,DRIVE)

2. Key in a (segment), and nnn (drive) separated by a comma

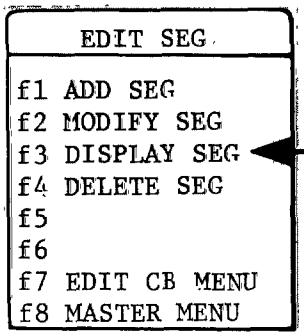
3. Depress Ⓔ.

Result:

Segment identification block appears with previously entered data. Refer to Figure 3-2 for the complete segment identification format. Use TAB key to move cursor to each of the modifiable fields. Key in data changes and/or additions. Depress Ⓔ when editing is completed.

If the syntax used in the modifiable fields is improper, an error message will appear and the segment identification may be re-edited to correct the error.

The abort key (f1) may be used to abort the modification if desired.



DISPLAY SEGMENT IDENTIFICATION

Function: This command is used to display previously entered segment identification information on CRT screen.

Format:

f3 (DISPLAY SEGMENT) a (segment), nnn (drive) Ⓔ

Procedure:

1. Depress f3(DISPLAY SEG)key on EDIT SEG menu.

   The following prompt will appear:

   ENTER SEG (,DRIVE)

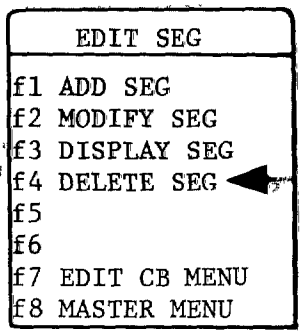
2. Key in a (segment), and nnn (drive) separated by a comma.

3. Depress Ⓔ.

Result:

Segment identification block appears.

(See segment identification format in Figure 3-2).



DELETE SEGMENT (Also see DELETE DRIVE)

Function: This command is used to delete a segment from a specified drive. All blocks within that segment will also be deleted. This command is also used for deleting drives.

Format:

f4 (DELETE SEG) nnn (drive), a (segment) (E)

Procedure;

1. Depress f4(DELETE SEG)key on EDIT SEG menu.

   The following prompt will appear:

   ENTER DRIVE (,SEG)

2. Key in nnn (drive) and a (segment) separated by a comma.

3. Depress (E).

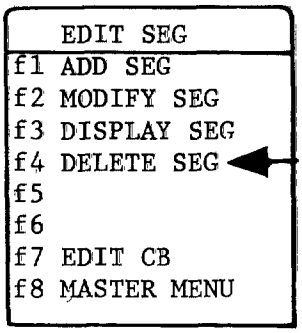
   The following prompt will appear:

   DELETE DRIVE NNN/SEGMENT A Y/N?

Result:

Y - should be keyed in if you wish to delete the specified segment; then depress (E).

N - should be keyed in if you wish to retain the specified segment; then depress (E). The segment will remain in the control system.



DELETE DRIVE (Also see DELETE SEGMENT)

Function: This command is used to delete a drive from a specified CBL system. All segments and their associated control blocks will also be deleted. This command is also used for deleting single segments.

Format:

f4 (DELETE SEG) nnn (drive) (E)

Procedure:

1. Depress f4 DELETE SEG key on EDIT SEG menu.

   The following prompt will appear:

   ENTER DRIVE(,SEG)

2. Key in nnn (drive).

3. Depress (E).

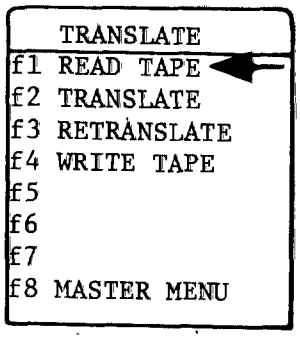
   The following prompt will appear:

   DELETE DRIVE NNN Y/N?

Result:

Y - should be keyed in if you wish to delete the specified drive; then depress (E). The drive and associated segments will be deleted from the control system.

N - should be keyed in if you wish to retain the specified drive; then depress (E). The drive will remain in the control system.



READ TAPE

Function: This command communicates the PMC hardware and software configuration to the FSS. This information serves as a basis for generating executable code after each translation.

The basic PMC configuration must be input to the FSS before a translation can be performed. The information is retained on a controller image (CI) and/or a partition (PART) tape.

Format:

f1 (READ TAPE) B (both tape drives),nnnn (system ID) (E)
OR
f1 (READ TAPE) L (left tape drive),nnnn (system ID) (E)
OR
f1 (READ TAPE) R (right tape drive),nnnn (system ID) (E)

Procedure:

1. Depress f1 READ TAPE key on TRANSLATE menu.

   The following prompt will appear:

   ENTER TAPE, ID
   [ ]

2. Key in B (both) or L (left), or R (right) applicable to installed tape position. Key in comma and control system ID.

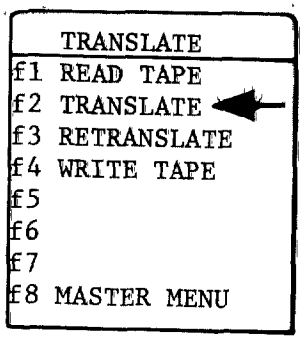
3. If L left or R right is keyed in, the following message appears:

   PLACE IMM/PART TAPE IN L/R UNIT. WHICH UNIT, L OR R?
   [ ]

   If the above message appears, insert tape, then key in L or R, then depress (E).

Result:

System nnnn configuration is read from the specified tape drive (both, left or right).



TRANSLATE

Function: This command converts perviously entered CBL blocks into a format recognized by the PMC. This translation can only be executed after a controller image (CI) tape and a partition (PART) tape are entered onto the FSS disc cartridge via the READ TAPE command. The CBL for the entire system is translated. Time required is proportional to size of system. After executing a TRANSLATE or RETRANSLATE function the operator must perform a SELECT command (in INITIALIZE menu) before selecting any other commands.

Format:

f2 TRANSLATE nnnn (system ID) (E)

Procedure:

1. Depress f2(TRANSLATE)key on TRANSLATE menu.

   The following prompt will appear:

   ENTER ID
   [ ]

2. Key in nnnn (system ID).

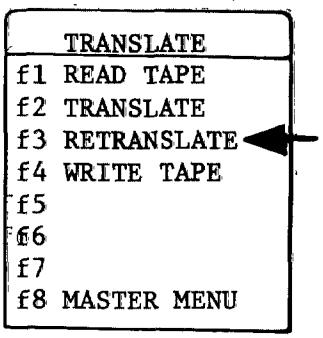
3. Depress (E).

Result:

During the translation process, CBL media for the nnnn system is translated. Passes 1, 2 and 3, and post processing messages appear on the CRT screen. Errors encountered are listed in the error file (EF).

Upon completion of the translation, other generated data is stored in various files.

A TRANSLATION COMPLETE message is displayed upon termination of the TRANSLATE function.



RETRANSLATE

Function: This command is used on those systems which were previously translated and where additional translations are required because of editing modifications to segments. This command significantly reduces translation time. After executing a TRANSLATE or RETRANSLATE function, the operator must perform a SELECT command (in INITIALIZE menu) before selecting any other commands.

Format:

f3(RETRANSLATE)nnnn(system ID)Ⓔ.

Procedure:

1. Depress f3(RETRANSLATE)key on TRANSLATE menu.

   The following prompt will appear:

   ENTER ID
   [ ]

2. Key in nnnn(system ID).

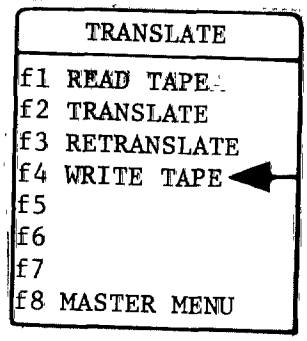
3. Depress Ⓔ.

Result:

During the retranslation process, CBL media for the nnnn system is retranslated. Passes 1, 2 and 3, and past processing messages appear on the CRT screen. Errors encountered are listed in error file(EF).

Upon completion of the retranslation, other generated data is stored in various files.

A TRANSLATION COMPLETE message is displayed upon termination of the retranslate function.



WRITE TAPE

Function: This command is used to initiate an output from the FSS disc cartridge onto the CI and the partition (PART) tapes, both of which are loadable into the PMC.

Format:

f4(WRITE TAPE)B (both tape drives), nnnn (drive) Ⓔ

OR

f4(WRITE TAPE)R (right tape drive), nnnn (drive) Ⓔ

OR

f4(WRITE TAPE)L (left tape drive), nnnn (drive) Ⓔ

Procedure:

1. Depress f4(WRITE TAPE)key on TRANSLATE menu.

   The following prompt will appear:

   ENTER TAPE, ID

2. Key in B (both), or R (right), or L (left), applicable to desired tape drive (s) and key in nnnn (system ID) separated by a comma.

3. Depress Ⓔ.

4. If R right or L left is keyed in, the following message will appear:

   PLACE IMM/PART TAPE IN L/R UNIT. WHICH UNIT, L OR R?

   If the above message appears, key in L or R; then depress Ⓔ.

Result:

PMC configuration files are recorded on FSS mini-cartridge tape.

PMC LOAD/COPY COMMANDS

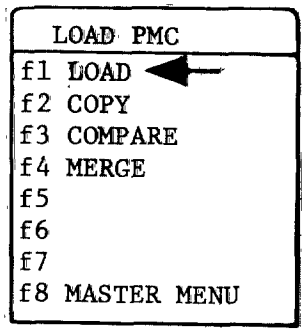
This group of commands is used to copy and compare the CBL load media to a cartridge tape. In each case, the PMC to be used must be connected to the FSS.

Load media is carried on cartridge tape. The tape drive number must be entered.

Load and copy functions applicable to an entire PMC memory can be specified. The compare function applicable to either a portion of the PMC memory or the entire PMC memory can be specified.

Loading of the PMC is restricted to the off-line mode.

Load media may be read from cartridge tape at the PMC terminal into the PMC. However, a direct down-line load from the FSS can also be performed. This bypasses the extra steps of writing a cartridge tape and then reading it into the PMC.



LOAD PMC

Function: This command is used to load the PMC from the FSS resident load media. The PMC must be down-line loaded (see Appendix J).

The control system ID must be entered with the command. If a merged control system (see MERGE PMC command) is desired to be loaded, the operator must specify the identification of this backup system as well as the identification of the originally translated control system.

Format:

f1(LOAD)nnnn(system ID)Ⓔ
OR
f1(LOAD)nnnn1 (system #1 ID),nnnn2(system #2 ID)Ⓔ

Procedure:

1. Depress f1(LOAD)key on LOAD PMC menu.

   The following display will appear:

   ENTER ID (,ID)
   [ ]

2. Key in nnnn1(system #1 ID)then depress Ⓔ.

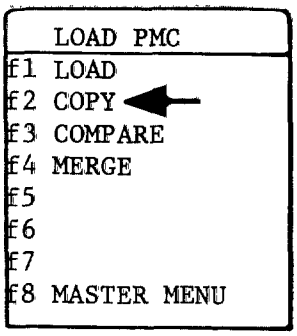
   OR, to load a merged system:

   key in nnnn 1 (system #1 ID)and nnnn 2(system #2 ID) separated by comma. Depress Ⓔ.

Result:

In the first case, system nnnn 1 was down-line loaded into the PMC.

In the second case, system nnnn 1 was previously merged to create system 2. System nnnn 2 was then down-line loaded into the same PMC as system nnnn 1 was in.



COPY PMC

Function: This command allows the PMC memory contents to be copied on an FSS cartridge tape in load media format. The PMC may be in any mode.

Format:

f2 (COPY)a(tape drive)Ⓔ

Procedure:

1. Depress f2(COPY)key on LOAD PMC menu.

   The following prompt will appear:

   ENTER TAPE
   [ ]

2. Key in alpha character B (both), or R (right), or L (left) tape drive.

3. Depress Ⓔ.

   Load media for the PMC's control system is read from the PMC and recorded on tape.

4. If only one tape was specified in step 2, the following prompt will appear:

   MOUNT IMM UTILITY TAPE IN CARTRDIGE -- TYPE L/R FOR PROPER UNIT
   ? [_]

   Mount a cartridge tape in the console's tape slot.

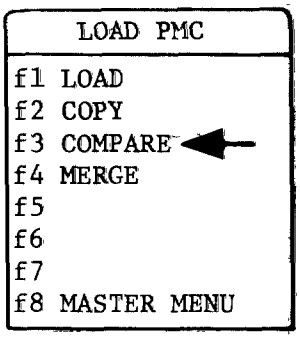
   Type in L (left) or R (right) tape.

   Depress Ⓔ.

Step 4 may have to be repeated until all data in the PMC memory has been copied onto cartridge tape.

Results:

The names of the PMC partitions are listed on the system CRT. Each of these partitions are copied onto the cartridge tape as a file. As each partition is copied, its ID will be listed on the CRT.



COMPARE PMC

Function: This command initiaties a comparison between the contents of a PMC and a cartridge tape load media image. The PMC may be in any operating mode.

Format: Three format options are available as follows:

Option (1) Compare complete system as follows:

f3(COMPARE)a(tape drive)Ⓔ

Option (2) Compare a specified area of data:

f3(COMPARE)a(tape drive),nnnnnn
(beginning address),nnnnnn (ending address) Ⓔ

Option (3) Compare specified partition.

f3(COMPARE)a(tape drive), aaaaa (PARTITION PROGRAM NAME) Ⓔ

Procedure:

1. Depress f3 COMPARE key on LOAD PMC menu.

   The following prompt will appear:

   ENTER TAPE, BEGIN ADDR, END ADDR

2. For option (1) key in R (right), or L (left), tape drive.

   For option (2) key in R (right), or L (left), tape drive nnnnnn (beginning address) nnnnnn (ending address) separated by commas.

   For option (3) key in R (right), or L (left), tape drive and PARTITION PROGRAM NAME* separated by a comma.

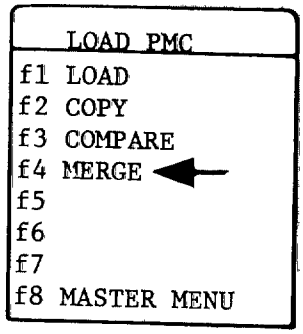
*PARTITION PROGRAM NAME refers to each of the partitions in the PMC such as IMM, FSSCM, CIAD, OUTPG, COMCM, MODE, PRIV1, PRIV2, etc.

3. Depress (E)

Result:

The contents of the PMC will be compared with specified cartridge tape load media.

Compare errors for each of the files are listed on the system CRT.



MERGE PMC

Function: This command allows the load files from a PMC system to be recorded on FSS disc cartridge from PMC memory (direct) or via cartridge tapes. The PMC system can be in any mode during the MERGE PMC operation. The new system ID must be different than the old one.

Format:

f4 (MERGE) a (tape drive) nnnn (old system ID), nnnn (new system ID) (E)

Procedure:

1. Depress f4(MERGE)key on LOAD PMC menu.

   The following prompt will appear:

   ENTER TAPE, ID, ID
   ?

2. Key in B (both), or R (right), or L (left), tape drive nnnn (old system ID), and nnnn (new system ID) (E) separated by 2 commas.

   OR

   Key in PMC, nnnn (old system ID), nnnn (new system ID) separated by 2 commas.

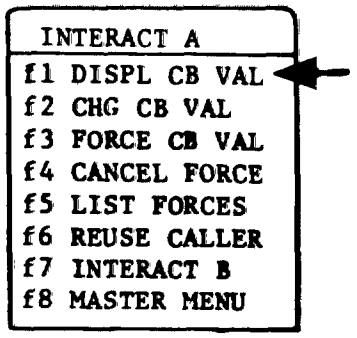
   If it is desired to merge the system from the PMC memory, enter "PMC" as the first parameter. If it is desired to merge the system from cartridge tapes, enter the tape drive designator that the tape is inserted in (L, R or B). These tapes must have been created through the COPY command in order to be properly formatted.

3. Depress (E).

4. The following message will appear:

```
DO YOU WISH TO VERIFY?  Y/N
?
```

To verify the tape, enter Y(E); otherwise, enter N(E).



DISPLAY CONTROL BLOCK WITH CURRENT VALUES

Function: This command displays a control block on the CRT screen with a corresponding value for each signal.

NOTE

The FSS must be connected to the PMC during the execution of all INTERACTION commands.

If the control block to be displayed is a reusable sequence, a CALLER command must be used first.

Format:

f1 (DISPL CB VAL) nnn (block number) a (segment), nnn (drive), nnnn (samples), nnnn (rate) (E)

Procedure:

1. Depress f1 (DISPL CB VAL) key on INTERACT A menu.

   The following prompt will appear:

   ENTER BLOCK# (,SEG(,DRIVE(,SAMPLES(,REPRATE))))

2. Key in nnn (block number), a (segment), nnn (drive), nnnn (number of samples), and nnnn (repetition rate).

   Parameter, if not entered, will default as follows:

   a. segment name to last one entered.
   b. drive number to the last one entered.
   c. #samples to one.
   d. rate to once per second.

The number of samples and repetition rate can be entered.

The number of samples specifies the number of times that the CRT screen will be refreshed with corresponding values for each signal. The repetition rate specifies the frequency that the CRT screen is refreshed.

If rate is entered as positive, the value is used as samples per second.

If rate is entered as negative, the value is used as seconds per sample.

3. Depress (E).

Results:

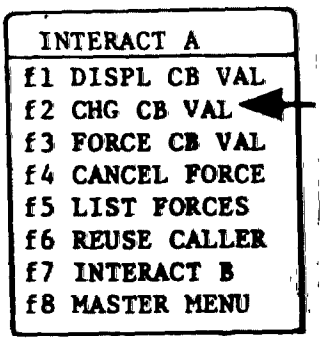
The specified control block will be displayed on the CRT. The values of each of the control block signals will be displayed next to the signal. If any input or output of the block is forced, the forced value will appear in half-bright inverse video.

DISPL CB VAL Restraints:

a. The effective repetition rate is restricted to the data transfer rate of peripheral devices. This will be on the order of 4 to 10 times per second.

b. Cannot display any block that has been modified since loading into the PMC.

c. Only regulating and sequencing blocks can be DISPL CB VAL'd with the exception of RCALL, RBEG, REND and LGIC blocks. Refer to the DIRECTO-MATIC ® Plus Control Block Language Users Manual (GEK-65426) for a definition of regulating and sequencing blocks.



CHANGE CONTROL BLOCK VALUE

Function: This command changes a signal value in the PMC to a new value. The command has an advantage over its Change Absolute (CHGABS) and Change Incremental (CHG INC) counterparts because it allows the user to change signals on a control block that does not have signal names. If the block to be changed is in a reusable sequence, the CALLER command must be used.

Format:

f2 (CHG CB VAL) nnn (block number), a (segment), nnn (drive) (E).

Procedure:

1. Depress f2 (CHG CB VAL) key on INTERACT A menu.

   The following prompt will appear:

   ENTER BLOCK # (,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

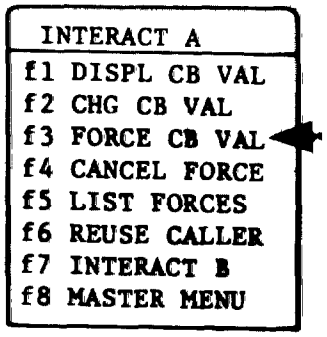
   A control block template with inverse video fields will be displayed for each changeable signal on the control block. Both inputs and outputs of the control block can be changed.

4. Key in any values desired into the inverse video fields. To change an input or output signal value , tab the cursor to the desired field and enter required value. Signal values are changed incrementally by preceeding the new value with a "%" sign. Any number of input or output fields may be changed using the same command.

5. After all data changes have been entered, depress (E).

Result:

All signal values keyed in will be inserted into the PMC.



FORCE/CANCEL SIGNALS

Function: This command is used to force signals and control block parameters in the PMC. It is also used to cancel forced values. A control block template is displayed with inverse video fields for each forceable signal on the control block. Up to sixteen input or output fields can be forced or cancelled using the same command. If the block to be forced is in a reusable sequence, the CALLER command must be used first.

Format:

f3 (FORCE CB VAL) nnn (block number), a (segment), nnn (drive) (E).

Procedure:

1. Depress f3 (FORCE CB VAL) key on INTERACT A menu.

   The following prompt will appear:

   ENTER BLOCK# (,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

   A control block template is displayed with inverse video field for each forceable signal on the control block. Both inputs and outputs of the control block can be forced.

4. Signals that are currently forced will appear in the inverse video fields. To force an input or output, tab the cursor to the desired field and enter the value to be forced. Currently forced values can be forced to another value. Forced signals can be cancelled by tabbing the cursor to the desired field and blanking out the forced value by use of the space bar.

5. Depress (E).

Results:

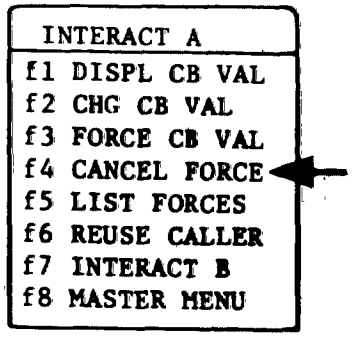
All forced values will be inserted into the PMC. Any previously forced values that were blanked out will be deleted from the PMC.

Force Restraints:

a. Any block modified since loading into the PMC cannot be forced.

b. Only regulating and sequencing control blocks can be forced with the exception of RCALL, RBEG, REND and LGIC. Refer to the DIRECTO-MATIC (R) Plus Control Block Language User's Manual (GEK-65426) for a definition of regulating and sequencing blocks.

c. If the output signal of a block in a reusable sequence is forced, that value is established for the specified user of that reusable sequence only.



CANCEL ALL FORCES

Function: This command is used to cancel forced values in an entire control system, drive, segment or block. If no portion of a system is specified, all currently forced values throughout the entire control system are cancelled.

Format:

f4 (CANCEL FORCE) nnn (drive), a (segment), nnn (block number) (E).

Procedure:

1. Depress f4 (CANCEL FORCE) key on INTERACT A menu.

   The following prompt will appear:

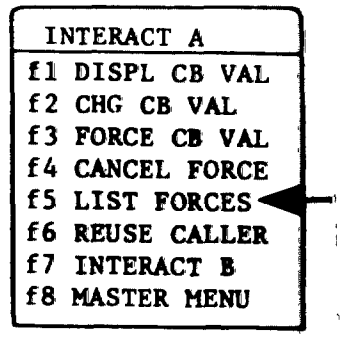
   ENTER (DRIVE(,SEG(,BLOCK#)))

2. Key in nnn (drive), a (segment), and nnn (block number) separated by 2 commas.

3. Depress (E).

Result:

The forced values cancelled by this command are listed on the system CRT. If there are no forces in effect for the specified drive, segment or block, a message is displayed on the CRT informing the operator as such.



LIST FORCED SIGNALS

Function: This command lists the identification and current value of all signals and parameters being forced in a control system.

Format:

f5 (LIST FORCES) (E).

Procedure:

1. Depress f5 (LIST FORCES) key on INTERACT A menu.
2. Depress (E).

Result:

A list similar to that shown below will appear on the CRT:

FORCE LIST

SEG ID BLK#	POS	USE	VALUE
350A090	3	I	6.83400E+1
350A200	1	I	T
350A210	5	O	32
230A010	6	O	1.05000E+1
100D088	1	I	F
100E010	2	I	O

If there are no currently forced values, a message on the CRT will indicate this fact.

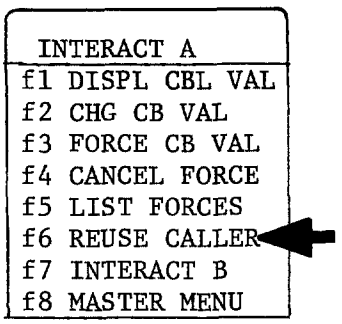
Definitions of force list parameters:

SEG ID BLOCK# - Fully qualified block number.

POS - Position in control block macro. (See CBL Reference Manual - GEK-65426).

USE - Input (I) or output (O) of the control block.

VALUE - Logic values true (T) and false (F), Count values, or analog values in scientific notation.



CALLER

Function: Before any signals in a reusable sequence can be displayed, recorded or changed, the RCALL control block that executes the reusable sequence must be identified. This can be accomplished by performing the CALLER command.

An attempt to reference a signal in a reusable sequence without executing a CALLER command will result in an error message.

Parameters are block number, segment and drive of which the segment and drive are optional.

Format:

f6 (REUSE CALLER) nnn (block number), a (segment), nnn (drive) (E).

Procedure:

1. Depress f6 (REUSE CALLER) key on INTERACT A menu.

   The following prompt will appear:

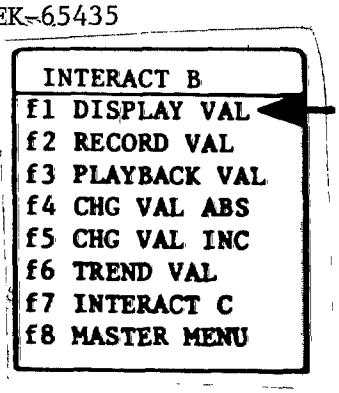
   ENTER (BLOCK #(,SEG(,DRIVE))

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

Result:

Signals within a reusable sequence may be displayed, recorded or changed.



DISPLAY VALUE

Function: This command allows the operator to display signal or data values from the PMC on the terminal CRT or graphics printer. Up to six signals or values can be displayed simultaneously. For each, a signal name and signal type must be entered via a block template. The number of samples and repetition rate can be specified. Both default to one. If the rate is entered as positive or fractional, the value is used as samples per second. If the rate is entered as negative, the value is used as seconds per sample.

Each value displayed is identified by one of two names as follows:

(1) A signal name, i.e., 350B110.
(2) An alphabetic name, i.e., SPEED.

Signal types recognized are: logic (L), count (C), analog (A) or text (T). One or more signal names and types must be input in each command.

If one or more signals specified are in a reusable sequence, they must all be in the same reusable sequence, and it must be identified by the CALLER command prior to using the DISPLAY VALUE command.

Format:

f1 (DISPLAY VAL) nnn (number of samples), nnn (rate) (E).

Procedure:

1. Depress f1 (DISPLAY VAL) key on INTERACT B menu.

   The following prompt will appear:

   ENTER (SAMPLES, REP RATE)

2. Key in nnnn (number of samples), and nnnn (repetition rate) separated by one comma.

3. Depress (E).

4. A block template will appear on the CRT. Key in the signal numbers or names and signal types that are to be displayed.

   For example:

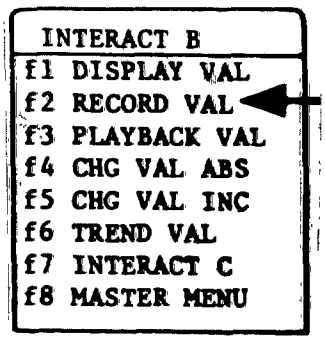
RECORD OR DISPLAY	
SIGNAL NAME	TYPE
3508110	A
STOP	L

Operator keys in signal names 350B110 and STOP, and signal types A (analog) and L (logical).

5. Depress (E).

Result:

Signals 350B110 and STOP are fetched from the PMC and displayed on the CRT the specified number of times at the specified rate. If the list device is set to the printer by the SET DEVICE command, the signal values will be listed on the printer instead of the CRT.



RECORD VALUE

Function: This command is used to record a signal or data values from the PMC onto the FSS disc memory. Up to six signals or values can be recorded simultaneously. The number of samples and repetition rate can be specified. Null entries will result in one sample and a rate of one sample per second. A positive and fractional rate is interpreted to be the number of samples per second. A negative rate is the number of seconds per sample.

Signal types recognized are: logic (L), count (C), analog (A) or text (T). One of more signal names and types must be input in each command.

Format:

f2 (RECORD VAL) nnnn (number of samples),nnnn (rate) (E).

Procedure:

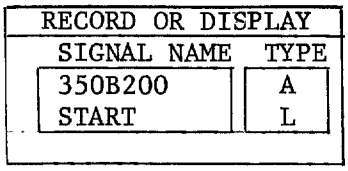
1. Depress f2 (RECORD VAL) key on INTERACT B menu.

   The following prompt will appear:

   ENTER (SAMPLES, REP RATE)

2. Key in nnnn (number of samples), and nnnn (rate) separated by one comma.
3. Depress (E).
4. A block template will appear on the CRT. Key in the signal numbers or names and signal types for display.

For example:

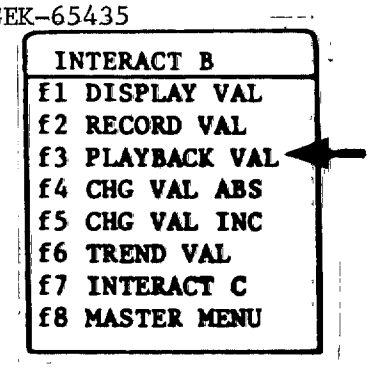


Operator keys in signal names 350B200 and START, and signal types A (analog) and L (logical).

5. Depress (E).

Result:

All signal values specified are recorded in a file at the specified rate. Use the PLAYBACK VALUES command to display the recorded values.



PLAYBACK VALUE

Function: This command allows the operator to retrieve and display signals or data values on the CRT or graphics printer. These items must have been previously recorded. The display repetition rate can be specified. Default is once per second. A positive or fractional entry is interpreted as the number of samples per second. A negative entry is seconds per sample. A group of samples may be selected by specifying the starting and ending sample numbers. The ending sample must be equal to or greater than the starting number.

Signal types recognized are: logic (L), count (C), analog (A) or text (T). One or more signal names and types must be input in each command.

Format:

f3 (PLAYBACK VAL) nnnn (rate), nnnn (starting sample number), nnnn (ending sample number) (E).

Procedure:

1. Depress f3 (PLAYBACK VAL) key on INTERACT B menu.

   The following prompt will appear:

   ENTER (REPRATE(,START SAMPLE(,END SAMPLE)))

2. Key in nnnn (rate), nnnn (starting sample number), and nnnn (ending sample number) separated by 2 commas.

3. Depress (E).

4. A block template will appear on the CRT. Key in signal numbers or names that are to be displayed.

For example:

PLAYBACK
SIGNAL NAME
350B110
STOP

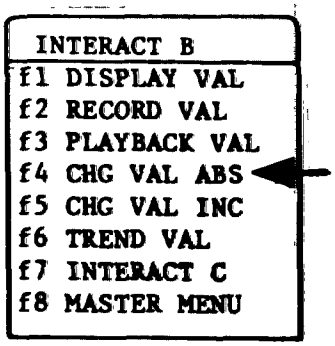
Operator entered signal number 350B110 and signal name STOP.

5. Depress (E).

Result:

The specified signal values, as recorded, are displayed on the CRT at the specific rate.

If the list device is set to the printer by the SET DEVICE command, the signal values will be listed on the printer instead of the CRT.



CHANGE VALUE ABSOLUTE

Function: This command allows a signal value in the PMC to be changed to a different value. Signal types that can be changed are analog (A) and count (C). If the signal to be changed is in a reusable sequence, the CALLER command must be used first.

Format:

f4 (CHG VAL ABS) nnnnnnnnnnnn (signal name or number), a (signal type), nnnnnnn (new value) (E).

Procedure:

1. Depress f4 (CHG VAL ABS) key on INTERACT B menu.

   The following prompt will appear:

   ENTER SIGNAL NAME, SIGNAL TYPE, VALUE

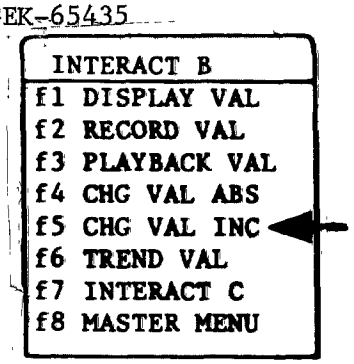
2. Key in nnnnnnnnnnnn (signal name or number),(a signal type), and nnnnnnn (new value).

3. Depress (E).

Result:

The old and new values appear on the CRT screen.

INTENTIONALLY LEFT BLANK



CHANGE VALUE INCREMENTAL

Function: This command allows a signal value in the PMC to be modified by some percentage value. A positive percent change is added to the current value while a negative percent change is subtracted from the current value. Signal types that can be changed are analog (A) and count (C). Text and logic signals cannot be changed incrementally. Any such attempt would result in undefined results. If the signal to be changed is in a reusable sequence, the CALLER command must be used first.

Format:

f5 (CHG VAL INC) nnnannn (signal name), a (signal type), nn (percent) Ⓔ.

Procedure:

1. Depress f5 (CHG VAL INC) key on INTERACT B menu.

   The following prompt will appear:

   ENTER SIGNAL NAME, SIGNAL TYPE, VALUE

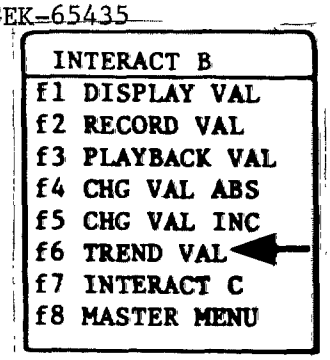
2. Key in nnnnnnnnnnnn (signal name or number) a (signal type), and nn (new value) separated by 2 commas.

3. Depress Ⓔ.

Result:

The old and new values will appear on the CRT screen.

INTENTIONALLY LEFT BLANK



TREND VALUE

Function: This command enables an output from the PMC which is picked up on a strip chart recorder. Up to twelve signals can be trended simultaneously. One command must be entered for each signal trended. The magnitude of the signal can be indicated by specifying a gain and an offset.

Output = G x S + O

Where: G = GAIN
S = SIGNAL
O = OFFSET

Default values of gain and offset are 1.0 and 0.0 respectively.

Trending on a channel is terminated by entering a command with a null or zero signal name. If the signal to be trended is in a reusable sequence, the CALLER command must be used first.

Format:

f6 (TREND VAL) n (channel) aaaa (name), a (type), nnnn (rate), nnnnn (gain), nnnnn (offset) (E).

Procedure:

1. Depress f6 (TREND VAL) key on INTERACT B menu.

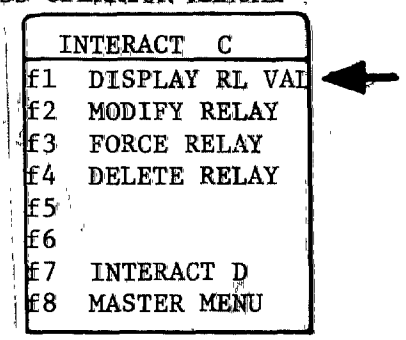
   The following prompt will appear:

   -- ENTER CHANNEL, SIGNAL NAME, SIGNAL TYPE (,REP RATE,VALUE, VALUE)

2. Key in n (channel), aaaa (name), a (type),nnnn (rate), nnnnn (gain), and nnnn (offset) separated by 4 commas.

3. Depress (E).

Result:
A signal from a specified PMC channel is trended on the strip chart recorder. Gain (G) and offset (O) calculations are described in the PMC/IMM Operators Manual (GEK-65442).



DISPLAY RELAY LADDER WITH VALUES

Function: This command results in a CRT display of a relay ladder diagram with signal names appearing adjacent to relay contacts. The number of samples and repetition rate default to 1 sample and once per second respectively. If the ladder to be displayed is in a reusable sequence, the CALLER command must be used first.

Format:

f1 (DISPLAY RL VAL) nnn (block number), a (segement), nnn (drive), nnnn (number of samples), and nnnn (rate) Ⓔ.

Procedure:

1. Depress f1 (DISPLAY RL VAL) key on INTERACT C menu.

   The following prompt will appear:

   ENTER BLOCK # (,SEG(,DRIVE(,SAMPLES, (,REP RATE))))

   [ ]

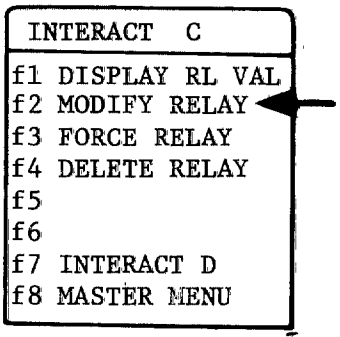
2. Key in nnn (block number), a (segment), nnn (drive), nnnn (number of samples), and nnnn (repetition rate) separated by 4 commas.

3. Depress Ⓔ.

Result: A designated relay ladder block will appear on the CRT. The values of each of the block's signals will be either open*(no inverse video) or closed (inverse video). Forced contacts will be indicated by half-bright inverse video fields on the signal name.

The display of values will continue for the specified number of samples at the specified rate.

* open = no current flow



MODIFY RELAY LADDER IN THE PMC

Function: This command modifies a relay ladder as it resides in the PMC. The command does not alter the relay ladder in the FSS. Segment and drive operands are optional and default to the last entered value.

If a modified ladder is accepted by the PMC, a prompt for the next command will appear. If the PMC rejects the modified ladder for some reason, an error message will appear on the CRT top line and the relay ladder, as modified, will be redisplayed along with the relay ladder commands menu. Correct the ladder and enter again.

Only signals previously defined in the control system may be added to the relay ladder using this command.

The segment that the relay ladder resides in must be disabled before the ladder can be modified. Refer to CHANGE MODE Command. The PMC PROGRAM MODIFICATION key switch must be set in the ENABLED position.

If the ladder to be modified is in a reusable sequence, the CALLER command must be used first.

Format: f2 (MODIFY RELAY) nnn (block number), a (segment), nnn (drive) (E).

Procedure: 1. Depress f2 (MODIFY RELAY) key on INTERACT C menu.

The following prompt will appear:

ENTER BLOCK # (,SEG (,DRIVE))

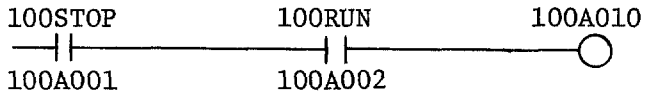
2. Key in nnn (block number) a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E)

Result: Example:

(Command) f2 10, A, 100

(CRT Display)



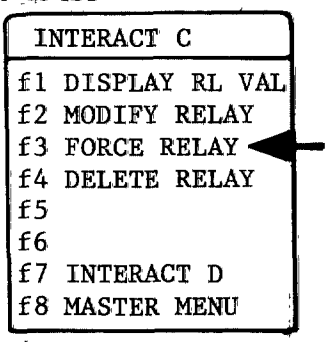
Relay ladder 100A010 will appear on the CRT. A menu of the relay ladder commands will appear in the lower right-hand corner of the CRT. Refer to the ADD CONTROL BLOCK command's section on relay ladder editing. If the modified ladder is accepted by the PMC, the operator will be prompted for the next command.

Constraint:

A segment cannot be modified in the PMC and FSS at the same time. For example, if the PMC operator is performing a MDFLD command on relay ladder 100A010 and the FSS operator attempts a MDFLD on a relay ladder in segment 100A, a message will appear:

"SEGMENT BEING MODIFIED IN PMC".

Wait for the PMC operator to complete the modify command, then try again.



FORCE RELAY LADDERS

Function: This command over-rides or forces relay ladder signals in the PMC. It also cancels forced values. Segment and drive operands are optional and default to the last entered value. If the ladder to be forced is in a reusable sequence, the CALLER command must be used first.

Format: f3 (FORCE RELAY) nnn (block number), a (segment), nnn (drive) (E).

Procedure:

1. Depress f3 (FORCE RELAY) key on INTERACT C menu.

   The following prompt will appear:

   ENTER BLOCK# (,SEG (DRIVE))

   [ ]

2. Key in nnn (block number), a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

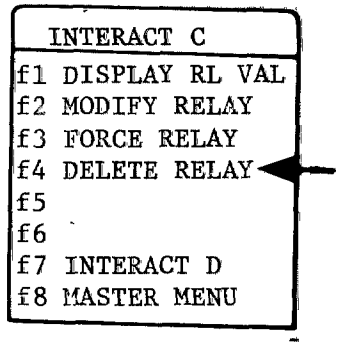
   The designated relay ladder block will appear with inverse video on the first forceable signal. Currently, forced contacts are indicated by half-bright inverse video fields over the signal name.

4. Move the cursor to the signal to be forced or cancelled. Type "F" to force the signal open or closed. The signal name will appear in half-bright inverse video. Type "T" to toggle forced values open or closed. To cancel a forced signal, type "C".

5. After forcing or cancelling the desired contacts, depress (E).

Result: If all forces or cancellations are accepted by the PMC, a prompt for the next command will appear on the CRT.

If the PMC rejects a force value for some reason (e.g., force list is full), a message will appear on the CRT top line and the relay ladder, along with currently forced values will reappear on the CRT.



DELETE RELAY LADDER FROM THE PMC

Function: This command deletes a relay ladder from the PMC. It does not delete the relay ladder from the FSS data base.

Format: f4 (DELETE RELAY) nnn (block number), a (segment), nnn (drive) (E)

Procedure:

1. Depress f4 (DELETE RELAY) key on INTERACT C menu.

   The following prompt will appear:

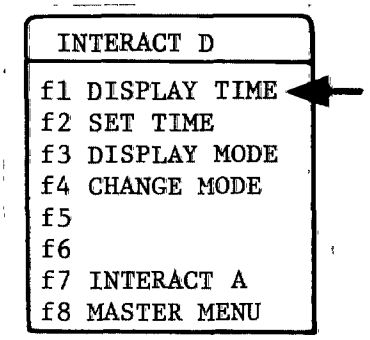
   ENTER BLOCK # (,SEG (,DRIVE))

   [ ]

2. Key in nnn (block number) a (segment), and nnn (drive) separated by 2 commas.

3. Depress (E).

Result: The designated relay ladder block will be deleted from the PMC.



DISPLAY TIME & DATE

Function: This command fetches current time and date from the PMC and displays them on the FSS CRT.

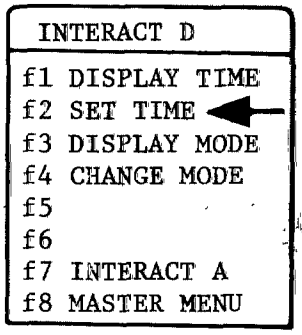
Format:

f1 (DISPLAY TIME)

Procedure:

1. Depress f1 (DISPLAY TIME) key on INTERACT D menu.

Result: The time and date will appear on the terminal CRT.



SET TIME & DATE

Function: This command is used to enter system time and date (optionally) into the PMC.

Format:

f2 (SET TIME) nn (hour), nn (minute), nn (second) (E)

OR

f2 (SET TIME) nn (hour), nn (minute), nn (second), nn (day), aaa (month), nnnn (year) (E)

Procedure:

1. Depress f2 (SET TIME) key on INTERACT D menu.

   The following prompt will appear:

   ENTER (HOURS, MINUTES, SECONDS (,DAY, MONTH, YEAR)

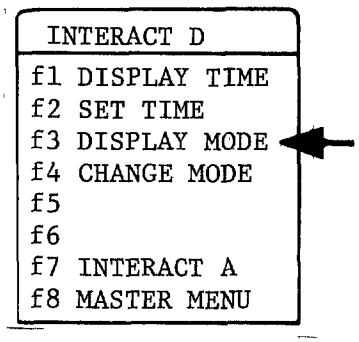
   

2. If it is desired to set the time in the PMC equal to the time in the FSS, depress (E).

3. If it is desired to set the timer, key in nn, (hour) nn,(minute) and nn (second) separated by 2 commas. Depress (E)

4. Or, if time and date are desired, key in nn (hour), nn (minute), nn (second), nn (day), aaa (month), nnnn (year) separated by 5 commas. Depress (E)

Result: The time in the PMC is set to the specified time.



DISPLAY MODE

Function: This command displays PMC system mode and, optionally, segment control mode.

There are three PMC modes:
(1) OFF-LINE
(2) STAND BY
(3) ON-LINE

There are two segment modes:
(1) DISABLED (D)
(2) ENABLED (E)

Command options are provided for various mode displays:

(1) System mode plus modes of all segments in system.
(2) System mode plus modes of all segments of a single drive.
(3) System mode plus the mode of a single segment.

Format:

f3 (DISPLAY MODE) (E)

OR

f3 (DISPLAY MODE) nnn (drive) (E)

OR

f3 (DISPLAY MODE) nnn (drive), a(segment) (E)

Procedure:

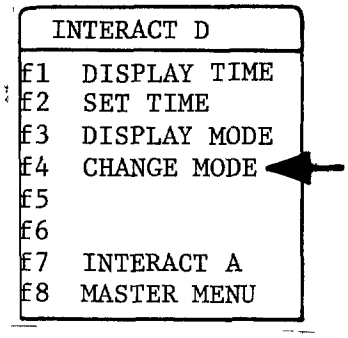
1. Depress f3 (DISPLAY MODE) key on the INTERACT D menu.

The following prompt will appear:

ENTER (DRIVE (,SEG))

2. Key in nnn (drive),and a (segment) separated by a comma.
3. Depress (E).

Result: The requested status appears on the CRT.



CHANGE MODE

Function: This command is used to change the mode of a specified segment.

The segment can be enabled or disabled by entering the code "E" or "D" respectively.

Format: f4 (CHANGE MODE) nnn (drive), a (segment), a (mode) (E)

Procedure:

1. Depress f4 (CHANGE MODE) key on the INTERACT D menu.

   The following prompt will appear:

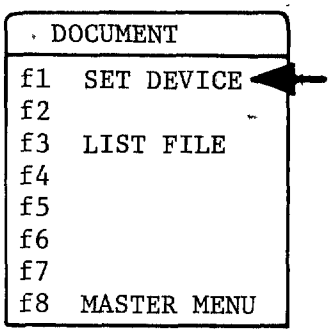
   ENTER DRIVE, SEG, MODE

   [ ]

2. Key in nnn (drive), a (segment), a (mode) separated by 2 commas.

3. Depress (E).

Result: The mode of the designated segment will be changed to the specified state.



SET DISPLAY DEVICE

Function: This command is used to route certain commands to a designated output device (graphics printer or CRT). The commands are:

DISPLAY CB
DISPLAY ID
DISPLAY VAL
PLAYBACK VAL

The designated graphics printer or CRT remains in effect until another SET DISPLAY DEVICE command is issued.

Format:

f1 (SET DEVICE) P (graphics printer) (E)

OR

f1 (SET DEVICE) C (CRT) (E)

Procedure:

1. Depress f1 (SET DEVICE) key on DOCUMENT menu.

The following prompt will appear:

ENTER DEVICE

2. Key in P (graphics printer) or C (CRT)

3. Depress (E).

Results: The output of DISPLAY CB, DISPLAY ID, DISPLAY VAL or PLAYBACK VAL command will appear on the designated device.

NOTE

IT IS RECOMMENDED THAT THE SYSTEM CRT CONSOLE TO WHICH THE PRINTER IS CONNECTED AS A SUBCHANNEL, BE PLACED IN UNBUFFERED MODE (*EQ, EQ#, UN) SO THAT NO OUTPUT TO THE PRINTER WILL WAIT ON THE COMMAND ENTRY LINE READ OF THE CRT. THIS IS ACCOMPLISHED AS FOLLOWS:

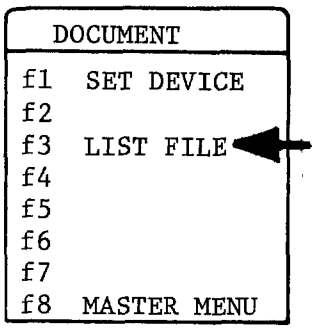
SET DISPLAY DEVICE (continued)

1) Shut down FSS.

2) Display the system prompt (*) on the terminal by depressing any key.

3) Key in *EQ, EQ#, UN (where EQ# is the equipment table number of the system console) and (CR) .

(INTENTIONALLY LEFT BLANK)



LIST FILE

Function: During editing and translation of a CBL system, several tables and lists are generated. Each of these files can be accessed and displayed on the CRT screen by use of the LIST FILE command.

Format:

f3 (LIST FILE) nnn (page), aa (file name), nnnn (System ID) Ⓔ

Procedure:

1. Depress f3 (LIST FILE) key on DOCUMENT menu.

   The following prompt will appear:

   ENTER PAGE (,FILE(, ID))

   [ ]

2. Key in nnn (page number), aa (file name), nnnn (system ID) separated by 2 commas.

3. Depress Ⓔ.

Result: The requested file is displayed on the CRT screen.

*Files which can be accessed are:

EF - Translation error file
PO - Hexadecimal listing of the load media (CI and partition tape data). (P-Zero)
SP - Schedule of continuous segments by priority
TC - Schedule of continuous segments by time count
PS - Procedural segments list
IS - Initialize segments list
VR - Signal cross reference list
IR - Input signal cross reference
OR - Output signal cross reference
IC - Internal signal cross reference
SL - System index list of segments, signals, etc.

4. TYPICAL EDITING AND TRANSLATING SESSIONS

USING THE FSS

The following editing sessions consists of the functions indicated below:

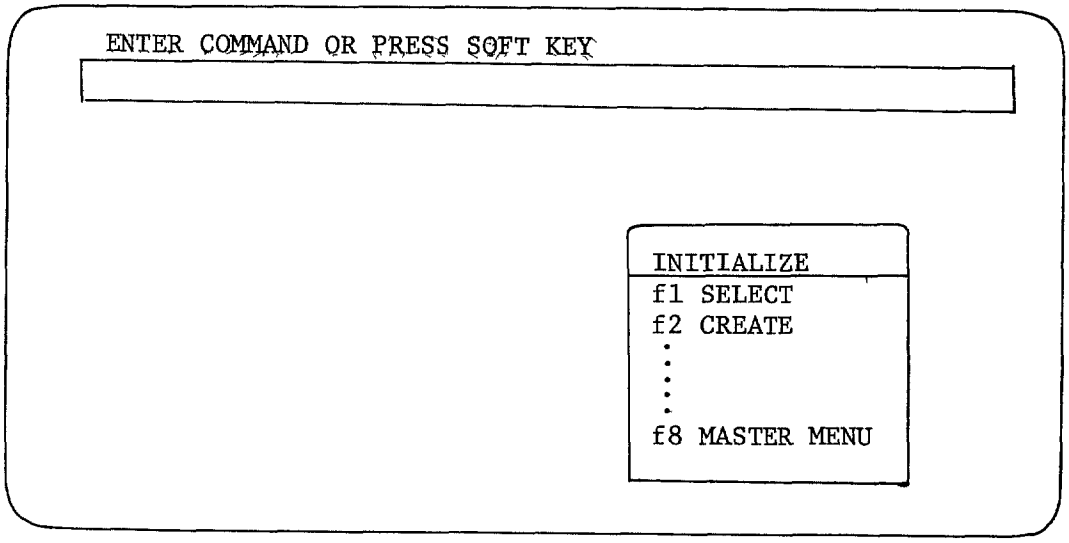
1. Creating a new control system.
2. Adding a segment to the new control system.
3. Adding some control blocks in the new segment.
4. Adding a second segment to the control system.
5. Adding some control blocks in the second segment.
6. Modifying a block in the first segment.
7. Translating the control system.
8. Performing a PMC interaction.

NOTE

The following editing sessions must be performed sequentially to obtain specified results.

A. CREATING A NEW CONTROL SYSTEM

1. Assume the INITIALIZE menu is displayed.



2. Depress f2 (CREATE) key. (CREATE will blink on menu).

3. A prompt ENTER ID will appear above the inverse video field; enter a 4-digit system number.

   Example: 7777 (E)

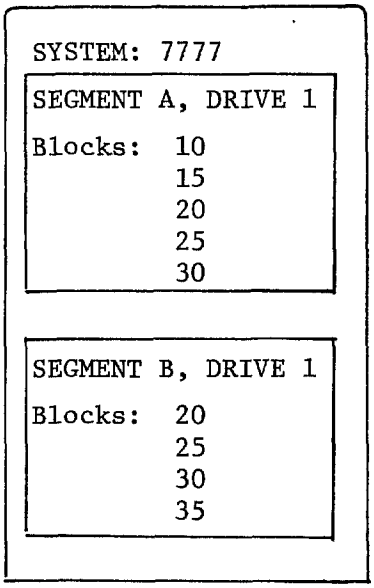
4. If control system 7777 had been previously entered, the FSS will respond with the prompt IT IS OK TO DELETE FILES Y/N?

5. Depress Y (E) if you wish to create a new system. Depress N (E) if you wish to abort the CREATE command. (In the latter case, perform a SELECT command instead of a CREATE.)

6. The command is processed as indicated by the inverse video field becoming dark. Upon completion of this process the inverse video field will reappear and CREATE will stop blinking.

NOTE

The system created above provides a vehicle to accommodate a number of segments, drives, control blocks, etc., as shown below and described in subsequent instructions.



B. ADDING A SEGMENT TO THE NEW CONTROL SYSTEM.

1. Depress f8 (MASTER menu) key.

2. Depress f2 (EDIT) key.

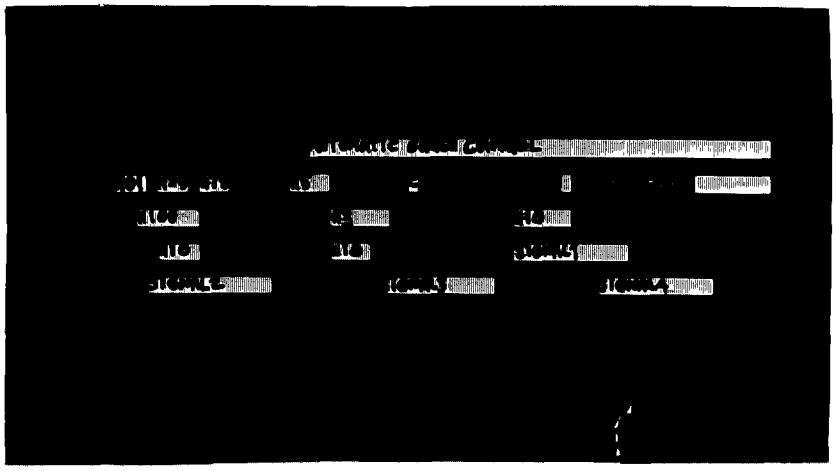
3. Depress f7 (EDIT SG) key.

4. Depress f1 (ADD SEG) key; ADD SEG on the menu will blink.

5. A prompt for SEG (,DRIVE) will appear. Enter segment and drive identification (if drive identification for this segment was not entered previously).

   EXAMPLE: A,1 followed by (E)

   A SEGMENT IDENTIFICATION block with numerous inverse video fields will appear. Mandatory entries are: TYPE, SGABLE, INABLE, RNABLE, DTABLE.



6. Use TAB key to sequentially move cursor to each of the fields. Enter required data into the fields.

7. Depress (E) and the prompt ENTER COMMAND OR SOFT KEY will appear.

C. ADDING SOME CONTROL BLOCKS IN THE SEGMENT.

1. Depress f7 (EDIT CB) key.

2. Depress f1 (ADD CB) key.

3. The following prompt will appear:

ENTER BLOCK TYPE, NUMBER (,SHEET(,SEGMENT(,DRIVE)))

Sample entry:

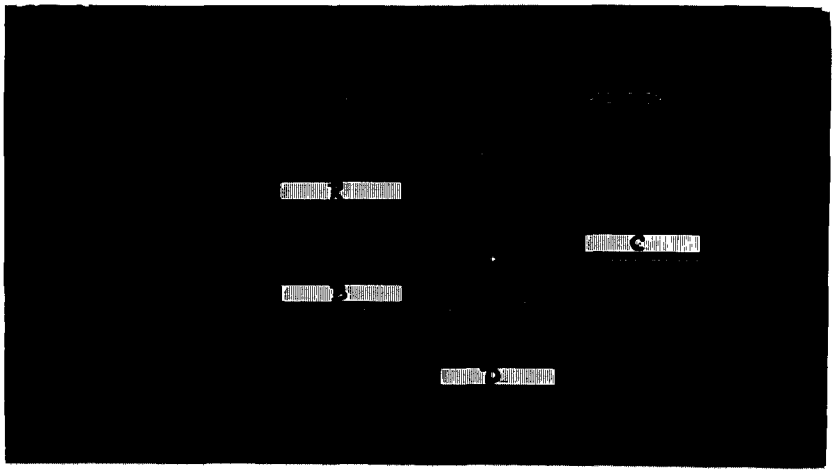
CADD, 10, S

4. Depress (E).

NOTE

Refer to CBL Reference Manual (GEK-65426) for types and characteristics of control blocks.

5. Enter applicable data into inverse video fields as shown in typical example below:



6. Depress (E).

7. Repeat steps 2 thru 6 to make additional control blocks.

D. ADDING A SECOND SEGMENT TO THE CONTROL SYSTEM

1. Depress f7 (EDIT SEG) key.

2. Depress f1 (ADD SEG) key.

3. The following prompt will appear:

   SEGMENT (,DRIVE)

   Sample entry:

   B

   (Drive #1 assumed for each drive but newly created drives must be indicated.)

4. Depress (E).

   A SEGMENT IDENTIFICATION block (appearing on sheet 4-3) with numerous inverse video fields will appear. Mandatory entries are: TYPE, SGABLE, INABLE, RNABLE, DTABLE.

5. Use TAB key to sequentially move cursor to each of the fields. Enter required data into the fields.

6. Depress (E) and the prompt ENTER COMMAND OR SOFT KEY will appear.

7. Repeat steps 2 thru 6 to make additional segments.

E. ADDING SOME CONTROL BLOCKS IN THE SECOND SEGMENT.

1. Depres f7 (EDIT CB) key.

2. Depress f1 (ADD CB) key.

3. The following prompt will appear:

   BLOCK TYPE, NUMBER (, SHEET(,SEGMENT(,DRIVE)))

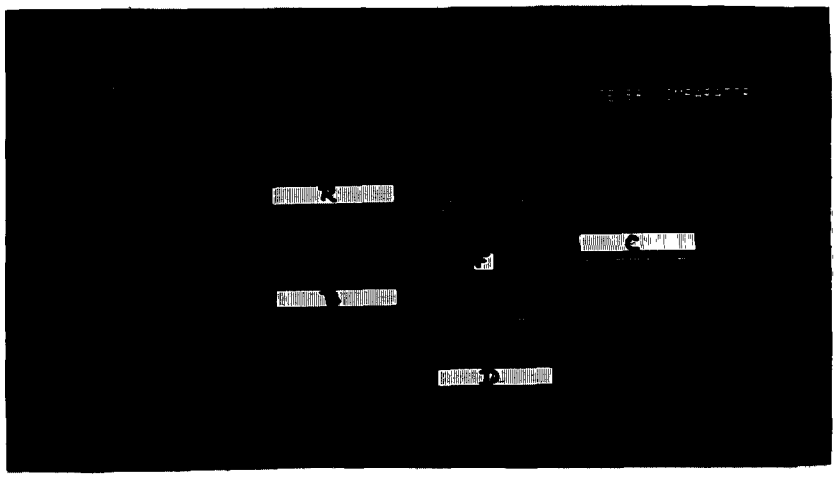
   Sample entry:

   LAND,15

   (Sheet S, segment B, drive 1 assumed if same as segment A.)

4. Depress (E) to obtain requested control block.

5. Enter applicable data into inverse video fields as shown in typical example below:



6. Depress (E).

7. Repeat steps 2 thru 6 to make additional control blocks in the second segment.

F. MODIFYING A CONTROL BLOCK IN THE FIRST SEGMENT.

1. Depress f2 (MODIFY CB) key.

   The following prompt will appear:

   ENTER BLOCK # (,SEGMENT(,DRIVE))

   Sample entry:

   10,A

   (Drive 1 assumed since it was previously used in other control blocks/segments.)

2. Depress (E) to obtain requested control block.

3. Modify data in inverse video fields by depressing TAB key to move cursor to fields of interest and replacing existing data.

4. Depress (E).

5. Repeat steps 2 thru 5 to modify additional control blocks.

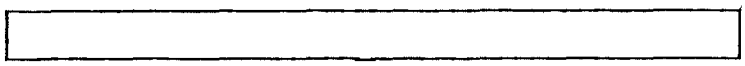
G. TRANSLATING THE CONTROL SYSTEM.

1. Depress f8 (MASTER MENU) key.

2. Depress f3 (TRANSLATE) key.

3. Depress f2 (TRANSLATE) key.

4. The following prompt will appear:
   ENTER ID

   Sample entry:

5. Depress Ⓔ.

   Approximately 5 minutes of execution time for the above example is required after the Ⓔ key is depressed.

NOTE

Upon completion of the above translation, depress f8 (MASTER MENU), f1 (INITIALIZE), then f1 (SELECT) to recall translated system. In above sample, 7777 Ⓔ would be entered.

H. PERFORMING A PMC INTERACTION

1. Depress f8 (MASTER MENU) key.

2. On the PMC front panel, perform the down line load procedure sequentially as described in Appendix I.

   The display lights should begin to cycle. When the display becomes 0 (all lights off) perform step 3.

3. Depress f4 (LOAD_PMC) key.

4. Depress f1 (LOAD) key.

5. A prompt for ID will appear. Enter applicable system ID number.

   Example: 7777

6. Depress (E) key and wait one minute for processing. The completion of processing is indicated by the appearance of the inverse video field at the top of the FSS CRT.

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5. APPENDICES

APPENDIX A. TERM DESCRIPTIONS

1. Signal numbers

- Signal numbers have the following format:

DDDSBBB where:

D - drive number (001-999)
S - segment number (A-Z, excluding O and I)
B - block number (001-999)

- There is a one-to-one correspondence with continuous blocks (from which they are defined).

- Can be used as the input to other continuous blocks.

- Can be used as an argument in a procedural block calculation.

Examples

100A200
5B373
7X002
998Z001

2. Signal names

- They can have a mixture of alphabetic and numeric characters up to a total of 12.

- They must begin with at least two alphabetic characters.

- They represent the synonym of a signal number.

- They are always associated with the output of a continuous block and its signal number.

- If the synonym contains four or less characters, it is local to the drive. With five or more characters, it becomes a global name.

- They can be used at the output of a continuous block (where they are defined).

- They can be used at the inputs to other continuous blocks.

- They can appear as an argument in a procedural block calculation.

Examples

Volts	Amp
Current	AA10077
AB68	100AFRED

3. Scalars

- Scalars are defined in an explicit fashion or in an implicit fashion.

- A Scalar is defined explicitly by appearing in the name field of a CNST or an INZS block.

- An explicitly defined Scalar may not be redefined; that is, it may only appear once in a system within a CNST or an INZV block.

- An explicitly defined Scalar may appear as the result argument of a procedural block. In this event, the result of the procedural calculation will be stored into the Scalar.

- A Scalar is defined implicitly by appearing as the result argument of a procedural block and by not being explicitly defined in a CNST or an INZS block.

- A Scalar may appear implicitly as the input argument of a continuous block many times.

- If a Scalar is defined explicitly, then the order of the block number where it is defined and other block numbers where it appears as a result argument should not matter.

- Can be a combination of alphabetic and numeric characters up to a total of twelve.

- They must begin with at least two alphabetic characters.

- They must be preceded by a special symbol; namely, #.

- They can appear as the input to continuous block, an argument of a procedural block or the result of a procedural block calculation.

- These names are always recognized globally despite their length.

Examples

#AB
#Current
#AB12
#Ampl

4. Array name

   - Comprised of a mixture of alphabetic and numeric characters up to a maximum of twelve.

   - Must begin with at least two alphabetic characters.

   - They must be defined by either an INZV or a DATV block.

   - These names are always recognized globally.

   - They can be used at the input to either a continuous or a procedural block.

   - They can appear as the result of a procedural block calculation.

   - They can be subscripted by a numeric or a two-alphabetic character index variable. (local synonym or scalar)

   Examples

Definition	Numeric Subscript	Variable Subscript
Amp	Amp (17)	Amp (IJ)
Current	Current (23)	Current (NN)
AB16	AB16 (6)	AB16 (PP)
Volt2	Volt2 (36)	Volt2 (#AB)

5. CSH address

- The format of a CSH address is as follows:

  /csgg/p where:

  c - channel (J, K or L)
  s - station (0-F)
  gg - group (00-7F)
  p - point (0-F, and blank for whole group. Add G if more than one bit and less than 16 bits are used.

- These names can only appear on the output of certain continuous blocks (i.e., LANO, LORO, COCB, AOCB, LORI, LAND, LCMP, LFFD, LFSR, LGIC, LSHT, LTDL, CCMP, CCTL and ACMP).

- These names can appear at the input to certain logic, count and analog blocks (i.e., AICB, CICB, ANSW, all logic blocks).

- These names cannot appear in the procedural blocks.

- These names are permitted as enable variables (unsubscripted).

Examples

/J37F/A
/K07F/G
/LF26/

6. Literals

- Comprised of numeric characters plus the possibility of the characters "+", "-", ".", and "E".

- Must always be preceded by the special symbol, #.

- They may be analog, count numbers, or logical.

- Scientific notation is allowed for analog numbers.

- The + operator will always be implied unless a - is used.

Examples

#31.25	#-6.237E5
#462	#-31.92E-4
#3.15E4	#16.4E-3

7. System globals

   - Can be comprised of numeric and alphabetic characters.
   - Must begin with a special symbol, &.
   - Can be used as inputs to continuous blocks and as arguments in procedural blocks.
   - Cannot ever be the output of any block.
   - Cannot be used in any special or pseudo block.

   Examples

   &MKL6 &T
   &TRUE &F
   &BIGC
   &PI
   &10X2

Summary

Referring to the aforementioned parameter types, we can now illustrate the compatibility of blocks with parameter types.

	PARAMETER TYPES						
	Signal Numbers	Synonyms	Scalars	Array Name	CSH Address	Literals	System Globals
Continuous Blocks							
Inputs	X	X	X	X	X	X	X
Outputs	X	X			X		
Procedural Blocks							
Inputs (amp)	X	X	X	X		X	X
Results	X	X	X	X			
Special Blocks							
INZS			X				
INZV				X			
INZO	X	X					
CNST			X				
DATV				X			

APPENDIX B. COMMAND OPERANDS

controller	- Control system identifier 0000 < NNNN < 7777 octal
drive	- Drive number. 001 ≤ NNN ≤ 999
seg	- Segment name. A to Z except I or O
blnb	- Control block sequence number. 001 ≤ NNN ≤ 999
sheet	- Elementary sheet identifier A to Z, except I or O
type	- Control block type mnemonic. AAAAA (4 or 5 alphabetic characters)
tape drive	- Left, right, or both cartridge tape drive(s). L, R or B
pin	- Signal pin number 1 ≤ N ≤ 9 Refer to the Control Block Language Reference Manual. (GEK-65426).
# samples	- Number of samples 1 ≤ NNNN < 1000
rate	- Sampling rate, samples per second -10000 ≤ NNNN ≤ 10000
name n	- Signal name, drive, segment, block. DDDSNNN or Signal name, alphabetic. AAAAAAA or DDDAAAA
channel	- Trending recorder channel. 1 ≤ N ≤ 6
gain	- Ranging gain for trending. 0.001 ≤ NNNNN ≤ 99999
offset	- Ranging offset for trending. -99999 ≤ NNNNN ≤ 99999
value	- New data value. NNNNNNN

use	- where in the block it is used I = input, O = output
hour	- Hour of the day. $00 \leq NN \leq 23$
minute	- Minute of the hour. $00 \leq NN \leq 59$
month	- Month of the year. Three letter abbreviation of month
day	- Day of the month. $01 \leq NN \leq 31$
year	- Current year NNNN
device code	- Code for graphics printer or CRT P = printer, C = CRT

APPENDIX C COMMANDS AND PARAMETERS

MNEMONIC	DEFINITION	PARAMETERS (Optional are enclosed in parentheses)
ADDCB	Add Control Block	type, blnb, (sheet), (seg), (drive)
ADDID	Add Segment Identification	seg, (drive)
CALLER	Reuseable Sequence Caller	(blnb), (seg), (drive)
CANCEL	Cancel all forces	(drive), (seg), (blnb)
CHGABS	Change Absolute	signal name, signal type, new value
CHANGE	Change by Control Block	blnb, (seg), (drive)
CHGINC	Change Incremental	signal name, signal type, percent
CMOD	Change Mode	drive, seg, mode
COMPAR	Compare PMC	tape drive, (begin address, end address) tape drive (partition program name)
COPY	Copy PMC	tape drive
DSPCB	Display CB/with Current Values	blnb, (seg), (drive),(# samples), (rate)
DELCB	Delete Control Block	blnb, (seg), (drive)
DELSD	Delete Segment/ Drive	drive, (seg)
DISPCB	Display Control Block	blnb, (seg), (drive)
DISPID	Display Segment ID	seg, (drive)
DISPV	Display Signal Values	# samples, rate
DLTLD	Delete Relay Ladder	blnb, (seg), (drive)
DMOD	Display Mode	(drive), (seg)
DSPLD	Display Relay Ladder/Values	blnb, (seg), (drive), (# samples) (rate)
DTIME	Display Time	-
FLOAD	Load PMC	controller, (backup controller)

MNEMONIC	DEFINITION	PARAMETERS (Optional are enclosed in parentheses)
FORCE	Force by Control Block	blnb, (seg), (drive)
FRCLD	Force Relay Ladder	blnb, (seg), (drive)
INIT	Initialize System	controller
LIST	List File	page, (file name), (controller)
LISTFR	List Force File	-
MDFLD	Modify Relay Ladder in the PMC	blnb, (seg), (drive)
MERGE	Merge PMC	tape drive, old controller, new controller PMC, old controller, new controller
MNTFD	Mount System	controller
MODCB	Modify Control Block	blnb, (seg), (drive)
MODID	Modify Segment Identification	seg, (drive)
PLAY	Playback Recorded Values	(rate), (starting sample), (ending sample)
RECORD	Record Signal Values	(# samples), (rate)
RETRAN	Retranslate	controller
RTAPE	Read PMC Configuration	tape drive, controller
SETDD	Set Display Device	device code
STIME	Set Time	(hours, minute, second), (day, month, year)
TRANSL	Translate	controller
TREND	Trend Signal Values	channel, name, type, (rate), (gain, offset)
WTAPE	Write Load Media	tape drive, controller

APPENDIX D AUDIT TRAILS

FSS can be run using audit trails for command entry. It allows the operator to automatically duplicate previous commands and screen entries. An audit trail is simply a file of FSS commands, parameters, and the CRT screen entries for each command.

Audit trails may be accessed in several manners. They can be written from operator input to commands, or by using a data extraction utility. They can be read in three modes -- automatic, semi-automatic, and manual.

To write the audit trail from operator input, start FSS using the command *WRITF or function key f2. Each command and screen input by the operator will be recorded in a file called AUDITO. This file can then be read in any desired mode to reproduce the same operator entries.

To save an existing system in the form of an audit trail, use the data extraction utility as follows:

1) Follow the normal shutdown procedure for FSS. (The terminal should be showing the ":" prompt.)

2) Enter the following command:
RU, DATAX

3) Respond to the SYSTEM ID? prompt with the 4 digit Control system ID desired. Terminate the input with a "return".

4) Respond to the HARDCOPY DESIRED (Y/N)? prompt with an "N" to receive status information on the terminal, or with a "Y" to print the status information on a printer.

5) The extraction utility allows for either the extraction of a single segment or an entire control system. To extract an entire system, press only "return". To extract a single segment, type the drive and segment (ex. 100A) followed by "return".

6) The extraction utility program will run, displaying or printing status information and segment summaries as each segment is completed. The created audit trail will contain a MNTFD command, followed by ADDID and ADDCB commands for segment and block in the control system or segment requested.

All the modes of reading audit trails -- automatic, semi-automatic, and manual -- read an existing audit trail. To start up FSS using one of these modes, enter one of the following commands in response to the ":" prompt:

automatic	*READA
semi-automatic	*READS
manual	*READM

APPENDIX E, MULTIPLE DATA BASE CONSIDERATIONS

The FSS is capable of supporting multiple control systems without swapping the disk unit. Each control system occupies two sections, or cartridges, on the disk.

Multiple system limitations are as follows. With a one 7906 disk system, 3 control systems can be supported. This factor increases by 4 for each 7906 disk unit that is added to the system. With 7905 disks, these figures are 2 and 3 respectively.

The assigned cartirdge numbers must correspond to the control system identification number (NNNN). Each control system's pair of cartridges must be assigned numbers NNNN and 1NNNN.

The PMC mode number must also correspond to the control system identification number. Multiple mode numbers may be assigned to the same physical PMC.

Files must reside on cartridges as follows.

CRnnnn	CR 1nnnn	
SInnnn	A( )nnnn	SBnnnn
SOnnnn	RCnnnn	SHnnnn
SNnnnn	SLnnnn	*COPY*
SFnnnn	ICnnnn	SCnnnn
SGnnnn	IRnnnn	
CInnnn	VRnnnn	
DSnnnn	SPnnnn	
	TCnnnn	
	ISnnnn	
CR 1Ø	PSnnnn	
	DInnnn	
MASTER	PØnnnn	
MISTER	DRnnnn	
CVTABS	VFnnnn	
ERRTXT	OCnnnn	
CVTABS	ODnnnn	
WELCOM	OPnnnn	
HELL1	EFnnnn	
FSSPAK	LFnnnn	
/FSS	OFnnnn	
\FSS	ORnnnn	
*WRITE	LMnnnn	
*READM	P1nnnn	
*READS	P2nnnn	
*READA	Q1nnnn	
FSSTR	QEnnnn	
HELLO	QHnnnn	
FSSSTP		
LSTEN2		
*FSS		

APPENDIX F GENERAL ERROR MESSAGES

Messages listed here are general in nature, arising during interpretation of any command input line.

ENTER COMMAND

Output on top line either alone or following another message indicates that another command can be entered.

COMMAND NOT IMPLEMENTED

Command name and operands are recognized but the command has not been implemented.

INVALID COMMAND NAME

Command name is incorrect. Check and re-enter the command.

OPERAND η INVALID, REENTER XXXXXXXXXXX

The η-th operand, named by XXXXXXXXXXX, is invalid. Enter a value for this operand only.

REQUIRED OPERAND η NOT ENTERED

A request for re-entry of the η-th operand of a command was made but no entry has been received. Re-enter the command or enter a new command.

OK TO IGNORE OPERAND η

Too many operands have been entered for the command. If the η-th operand is not necessary, enter YES or Y. If the η - th operand is required, enter NO or N.

COMMAND HAS NO OPERAND η

Operand η has been specified as necessary, but the command does not have an η-th operand. Re-enter the command or enter a new command.

OPERAND η IS IMPROPER TYPE

The η-th operand is required, but the entered value is of incorrect type. For example, an alphabetic is entered where a numeric is required. Re-enter the command or enter a new command.

OPERAND η INVALID IN CONTEXT

Although this command may use an η-th operand, in this case it is invalid in the context of other operands. Re-enter the command or enter a new command.

APPENDIX G TRANSLATOR ERRORS

Fatal Errors

-1 to -17	FMP Errors
-20	Executive service error
-21	Invalid parameters from command listener
-22	No source or master or type 7 object file under name given code for control system ID
-23	Checksum error in type 7 image file
-24	DBA file access error
-25	DBA invalid calling arguments
-26	#SSB's encountered exceeds max. # specified in KIB
-27	Size of SSB in KIB exceeds buffer size in ASMBL
-28	Error attempting to allocate SSB space
-29	# KIB pointer table entries out of limits
-30	# words in KIB data block out of limits
-31	Data Base coding error in object edit MDR
-32	-TBD- routine accessed
-33	Vector overrun
-34	Allocator - invalid base selected
-35	Allocator - variable type not allowed
-36	Allocator - invalid variable type
-37	Required symbol table entry not found
-38	Data allocated exceeds maximum memory size
-39	Segment ID block has no type
-40	DISP subroutine - Bad return
-41	SIOP - Global track error
-42	SIOP - Invalid record sequence in SH file
-43	P2 - Outbuffer overflow
-44	Not enough partitions in RTE
-45	Not enough memory for defined tasks
-46	Symbol table entry missing
-47	Could not schedule "FMGR"
-180	Symbol table space filled
-181	Invalid symbol operation code
-182	Add symbol table record - same name but not same type
-183	Invalid symbol table block number

TRANSLATOR ERRORS (continued)

Pass 1 Errors

Error #	Meaning
1	Undefined Signal
2	Undefined Symbol
3	Undefined Synonym
4	Invalid SNL Type Code
5	Undefined Segment ID Enable Variable
6	Undefined KIB Symbol $CSEGS
7	Undefined KIB Symbol $CSIZE
8	Undefined KIB Symbol $CADDR
9	Undefined KIB Symbol $CUSED
10	Undefined KIB Symbol $PSEGS
11	Undefined KIB Symbol $PSIZE
12	Undefined KIB Symbol $PADDR
13	Undefined KIB Symbol $PUSED
14	Undefined KIB Symbol $ISEGS
15	Undefined KIB Symbol $ISIZE
16	Undefined KIB Symbol $IADDR
17	Undefined KIB Symbol $IUSED
18	Undefined KIB Symbol $ZSEGS
19	Undefined KIB Symbol $ZSIZE
20	Undefined KIB Symbol $ZADDC
21	Undefined KIB Symbol $ZUSED
22	Undefined KIB Symbol $ZADDH
23	CSH Address Invalid
24	CSH Address Not Implemented
25	Illegal Double Word CSH Address
26	Intermediate File Buffer Overflow
27	CSH Address Conflict
28	Undefined table referenced
29	System global symbol not defined
30	System global symbol &F is missing
31	System global symbol &T is missing
32	System global wrong type
33	Undefined KIB Symbol $FREQ

Data Block Error Messages

400	Data block error	413	Symbol file update error
401	Invalid header column	414	$DBTBL symbol undefined
402	Duplicate block number	415	Data block size error
403	Invalid request flag name		
404	Request flag undefined		
405	Request flag must be logic		
406	Request flag already allocated		
407	Invalid signal name		
408	Signal undefined		
409	Signal already allocated		
410	Invalid input signal type		
411	Invalid output signal type		
412	Invalid data type		

TRANSLATOR ERRORS (continued)

Pass 2 Errors

Error #	Meaning
501	Invalid indexing in control block
502	Required entry omitted
503	Unable to load constant type
504	Integer value out of range
505	Unused forward pointer
506	Invalid Psuedo record type
507	TXTO/PTXTO - input signal lacks associated picture
508	TXTO/PTXTO - ill-formed picture found
509	TXTO/PTXTO - internal logic error within TXTO/PTXTO found
510	Invalid character in LGIC block
511	Variable type not logical in LGIC block
512	Number of parenthesis do not balance in LGIC block
513	Missing operator
514	Not enough parameters
515	Invalid variable block sub type
516	Undefined input to LGIC block
517	Data base object edit error
518	Undefined routine from data base

Procedure Segment/Block Errors

519	Submodule pointer without submodule
520	Missing PRETN
521	Missing PSUBM
522	Missing PMAIN/too many
523	Missing PMEND/too many
524	Mixed types
525	Illegal type
526	Ill-formed time expression (PACTV, PDELY)
527	Sizes not equal
528	Illegal operator
529	Block length > 31 or 62 words
534	Misuse of reuseable sequence signal
535	CSH address in reuseable sequence
536	Missing REND block
537	Missing RBEG block
538	RCALL in reuseable sequence
539	Reuseable sequence not defined
540	Illegal SSB address
541	Parameter supplied but not required
542	Parameter required but not supplied
543	Parameter type does not match RBEG
544	Illegal parameter type
545	Too many logical variables in sequence

Pass 3 Translator Errors

1000	Illegal Record Type
1001	Illegal base type (P2)
1002	Variable Base Type Offset gt allocated size (P2)
1003	Body offset greater than allocated size
1004	Illegal subtype
1005	Error record encountered
1006	Illegal base type (SF)
1007	Variable for type offset > allocated size

APPENDIX H. LOAD/COPY ERROR MESSAGES

CONTROLLER IDENTITY ERROR

Controller identification read from the PMC does not match identity entered in the command.

TAPE IDENTITY ERROR

Controller identification read from the cartridge tape Load Media does not match identity entered in the command.

Check that cartridge tape is the Load Media for the intended Controller.

CONTROLLER IS ON-LINE

PMC was found to be on-line when a system load was attempted. Place controller off-line before attempting the load again.

CONTROLLER IS STANDBY

PMC was found to be in standby mode when a system load was attempted. Place controller off-line before attempting the load again.

CONTROLLER DISCONNECTED

PMC was found to be in standby mode when a system load was attempted. Place controller off-line before attempting the load again.

DRIVE $\eta\eta\eta$ NOT FOUND

The request drive number "$\eta\eta\eta$" is not present. Check the entered drive number.

SEGMENT a NOT FOUND

The requested segment, number "a", is not present. Check the entered segment alphabetic.

TAPE READ ERROR, DRIVE η

A cartridge tape read error on drive number "η" has been encountered.

LOAD/COPY ERROR MESSAGES (Continued)

TAPE WRITE ERROR, DRIVE n

A cartridge tape write error on drive number "n" has been encountered.

CONTROLLER DATA LINK ERROR

An error was encountered in transferring data to or from the PMC.

APPENDIX J PMC DOWN LINE LOAD

Before the PMC can be loaded from the FSS by performing the FLOAD command, the PMC must be down line loaded. This can be accomplished at the PMC with the following procedure.

HALT
REGISTER SELECT S
CLEAR DISPLAY
ENTER 1 100 001 110 000 000 on the display
STORE
PRESET
IBL
CLEAR DISPLAY
ENTER Control System Indentification on the display in octal format.
STORE
PRESET
RUN

The display lights should begin to cycle. When the display becomes 0 (all lights off) an FLOAD command may be performed on the FSS.

APPENDIX K RELAY BLOCK EDITING ERROR MESSAGES

The relay editing error messages are of three types:

- (*1) - Invalid operator inputs
- (*2) - Transmission problems between the CPU and CRT due to one of the following:
  - bad cabling
  - loose connectors
  - terminal data - communication card malfunction
  - CPU hardware or software malfunction
- (*3) - CRT internal firmware of hardware malfunction.

RL ERROR 1 - TRANSMISSION ERROR (*2)

Data transmission problem between FSS computer and terminal

RL ERROR 2 - TRANSMIT TIME WARNING (*2)

This warning results from the terminal trying to transmit for 2.5 seconds and getting no response from the FSS computer. The terminal will try to transmit indefinitely.

RL ERROR 3 - DATA RECEPTION ERROR (*2)

Invalid data received from FSS computer.

RL ERROR 4 - RECEIVE TIME WARNING (*2)

This warning occurs when the terminal has waited for response from the CPU for longer than 2.5 seconds.

RL ERROR 5 - INPUT BUFFER OVERFLOW (*3)

Data from the CPU has overflowed the terminal input buffer.

RL ERROR 6 - LRC CHECK ERROR (*2)

This indicates bad transmission of data from the CPU.

RL ERROR 7 - FIRST RELAY NOT IN CORNER (*1)

The first relay rung must start in the upper left corner.

RL ERROR 8 - OPEN CONTACT ERROR (*1)

One end of a contact is not connected to a rung.

RL ERROR 9 - SHORT CIRCUIT ERROR (*1)

There is a short around a contact.

RL ERROR 10 - REVERSE CURRENT ERROR (*1)

A contact has been inserted such that current can pass through it in either direction.

RL ERROR 11 - SORT ERROR #1 (*3)

Internal CRT software error.

RL ERROR 12 - SORT ERROR #2 (*3)

Internal CRT software error.

RL ERROR 13 - SORT ERROR #3 (*3)

Internal CRT software error.

RL ERROR 14 - ILLEGAL MESSAGE NUMBER (*3)

Internal CRT software error.

RL ERROR 15 - INVALID INSTRUCTION OP CODE (*2)

An invalid instruction op-code was encountered during the process of generating a relay ladder display.

RL ERROR 16 - RELAY LADDER REQ. >5 ROWS (*1)

The relay ladder to be displayed will not fit in the 5 row format of the CRT. This error is caused by the operator editing a relay ladder into a configuration which, when "normalized", will result in a relay ladder that requires more than 5 rows. Correction of this problem requires the opeator to reconfigure the relay ladder into two KRLY control blocks.

RL ERROR 17 - RELAY LADDER REQ. >7 COLUMNS (*1)

The relay ladder, when normalized, requires more than 7 columns. Correction of this problem requires that the relay ladder be reconfigured into two KRLY blocks.

RL ERROR 18 - POPS NOT EQUAL PUSHES (*3)

This is an internal software error.

RL ERROR 19 - INVALID IMAGE DATA BASE OP CODE (*3)

Internal CRT software error.

APPENDIX L DATA BLOCK OPTION

In order for a user to have the ability to transfer blocks of I/O signals between the PMC and another CPU, the data block option may be used.

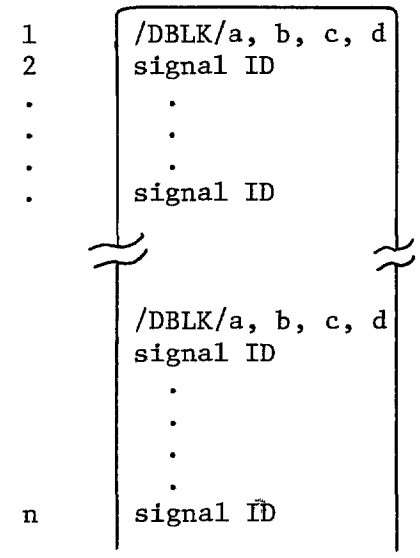
Data Block Structure

The user specifies the signals to be included in the data block transfers by building an ASCII file with all the necessary information. The file will be used by the FSS translator at translation time to build the data blocks which may be accessed by external sources.

The data file has the following format:

file ID = DBXXXX where XXXX = control system ID

Record #



where: a = Data Block # (1 to max of 16)

b = Block type
- I = input block; values are used as inputs to control blocks
- O = output block; values are outputs from control blocks

c = Value type
- A = Analog & Count
- L = Logical

APPENDIX L DATA BLOCK OPTION (continued)

d = Request flag; an internal logical signal which will be set when a data block access request is made by the external source. For an input block, it means there is a request for a Data Block.

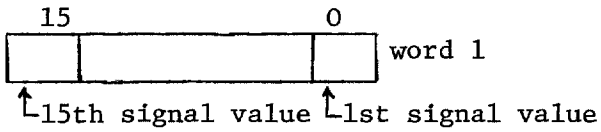
The maximum number of words per data block is 127.

Input block values must be scalers. Output block values must be signal numbers or signal names.

The values in Data Blocks have the following formats:

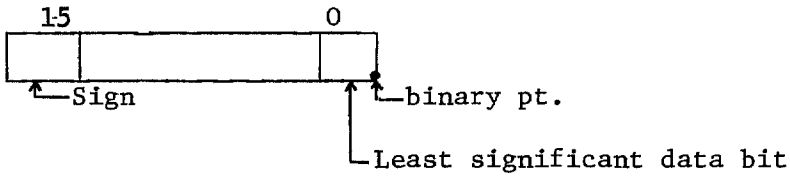
Logical

One bit representation beginning with bit 0 of the first word progressing to bit 15 and then to bit 0 of the second word, etc. The bits are assigned sequentially in the same order in which the signal IDs are encountered in the file.



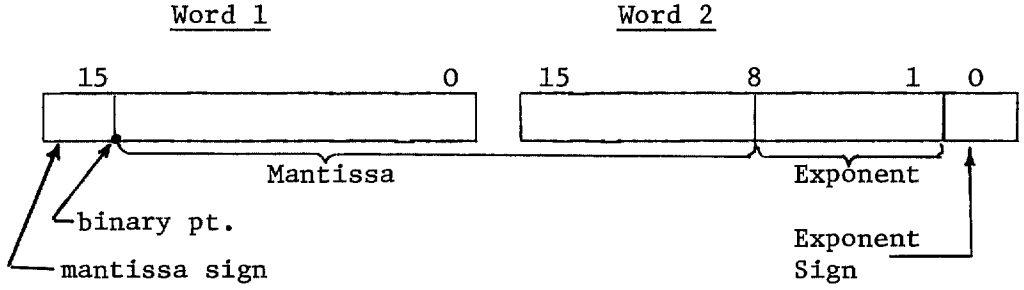
Count

Sixteen bit signed integers with negative numbers represented in two's complement form; assigned sequentially in the same order in which the signal IDs are encountered in the file.



Analog

Thirty-two bit floating point representation with negative mantissa or exponent represented in two's complement form; assigned sequentially in the same order in which the signal IDs are encountered in the file.



APPENDIX L DATA BLOCK OPTION (continued)

Example Data Block File:

```
FILE ID = DB1001

/DBLK/1, I, A DB1FG
#INP01
#INP02
#INP03
#INP04
/DBLK/2, O, A, DB2FG
100A020
200B050
300C110
SPEED
SPDREF
/DBLK/3, I, L, DB3FG
#LIN01
#LIN02
#LIN03
#LIN04
#LIN05
#LIN06
/DBLK/4, O, L, DB4FG
100A010
100A020
100A030
400C520
STRTF1
STOPF2
300JOG
400JOG
```

APPENDIX L DATA BLOCK OPTION (continued)

Restrictions

1) Only scaler signals (e.g., #SPEED) are allowed in input Data Blocks.

2) Only signal numbers (control block IDs) and signal names (synonyms for control block IDs) are allowed in output Data Blocks.

3) Only logical signals may appear in an "L" type Data Block.

4) Only analog or count signals may appear in an "A" type Data Block.

5) The "Request Flag" ID for each Data Block must be a logical signal. It may be a signal number or signal name.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

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